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Clickstream Data Analysis to Predict Online Purchase Behavior

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Statutory Declaration:

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Abstract

Online retailers need to comprehend and predict the purchasing habits of their customers, because e- commerce platforms are growing so quickly. Clickstream data offers important insights into customer intent by capturing fine-grained user interactions like page views, product exploration, and navigation sequences. Although machine learning models have demonstrated strong predictive performance regarding purchases, which restricts their applicability to regulatory compliance and business decision-making and helpful in coming up with business decisions or the law.

The purpose of this thesis is to determine whether interpretable machine learning models can accurately predict purchase intent from clickstream data at the session level while offering clear justifications for their predictions. A well-rounded and repeatable experimental pipeline includes data preprocessing, session-based feature engineering, model training, evaluation and explainability analysis. The LightGBM will be used as a fast ensemble method with SHAP-based explanations, and the Logistic Regression will be used as a decipherable baseline model.

ROC-AUC and Precision-Recall AUC are used as the models to measure model performance due to the discrepancies in the number of classes in a purchase data. Explainability is evaluated by Coefficient analysis, global feature importance, SHAP summary plots and local explanation visualizations. Although explainability methods give valuable insights into the behavioral motivators such as product variety, price fluctuations, and depth of browsing, experimental findings indicate that interpretable models have competitive predictability.

These findings indicate that interpretable machine learning models are suitable to be applied practically in e-commerce because they can be effectively used to complete purchase prediction tasks, preserving their transparency and trustworthiness.

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List of Abbreviations

Abbreviation	Full Form / Description
AI	Artificial Intelligence
API	Application Programming Interface
AUC	Area Under the Curve
CART	Classification and Regression Trees
CI	Confidence Interval
CSV	Comma-Separated Values
CTR	Click-Through Rate
DL	Deep Learning
EDA	Exploratory Data Analysis
FN	False Negative
FP	False Positive
GBM	Gradient Boosting Machine
IID	Independent and Identically Distributed
KPI	Key Performance Indicator
KDE	Kernel Density Estimation
LGBM	Light Gradient Boosting Machine
ML	Machine Learning
NLP	Natural Language Processing
OHE	One-Hot Encoding
PR-AUC	Precision–Recall Area Under Curve
ROC	Receiver Operating Characteristic
ROC-AUC	Receiver Operating Characteristic – Area Under Curve
SHAP	SHapley Additive exPlanations
SMOTE	Synthetic Minority Over-sampling Technique
TP	True Positive
TN	True Negative
XAI	Explainable Artificial Intelligence

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Chapter 1

Introduction

This chapter explains the history of using clickstream data analysis to forecast consumer behavior as well as the reasons for choosing the research topic for this thesis. It also gives a summary of the other chapters in this thesis.

1.1 Background

Online retailers need to comprehend and predict the purchasing habits of their customers, as e- commerce sites grow quickly. By keeping track of things like page visits, product searches, and navigation paths, clickstream data can give information about the preferences and intentions of your customers. But a lot of high-performing models are hard to understand, which means they aren't good for making business decisions or following the rules. Even though machine learning models are very good at predicting purchases, this is still true.

This thesis investigates the capability of interpretable machine learning models to accurately predict purchase intent from session-level clickstream data while providing transparent rationales for their predictions. A complete and reproducible experimental framework is established, including steps for preparing data, creating session-level features, training models, testing them, and looking into how to explain them. LightGBM is a high-performance ensemble method which is more compatible with SHAP-generated explanations. The Logistic Regression is an evident baseline model.

We examine the performance of the model on ROC-AUC and Precision-Recall AUC since the purchase data is not evenly distributed in terms of classes. Coefficient analysis and global feature importance can be used to assess explainability and you can run coefficient analysis, global feature importance, SHAP summary plots, and local explanations using visualization. The results of experiments indicate that interpretable models are quite effective in predictive ability. On the other hand, explainability methods produce relevant information on behavioral variables such as product variety, price variations as well as time taken by people to browse.

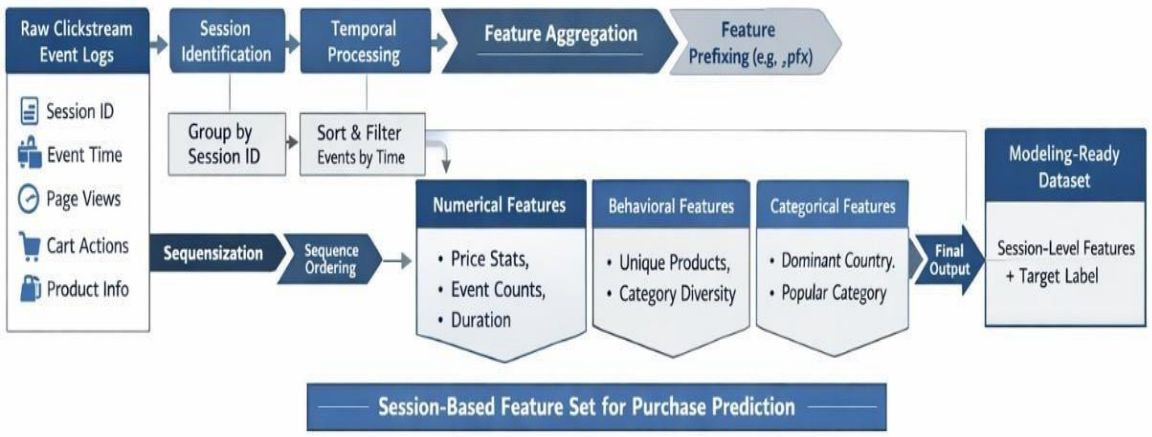


Figure 1.1: Transparent Modelling Approach

The growing accessibility of extensive behavioural data sets that you can use. This has brought about facilitation to employ machine learning and data mining to glean more profound understandings of user behaviour. Descriptive analytics are effective at illustrating past trends, but tend to overlook non-linear relationships that are very complex in data regarding user interactions with each other. The modern techniques to predictive modelling include logistic regression, tree based techniques and gradient boosting techniques and all assist business make models that can predict the likelihood of conversion given the activity of the users within a session.

Predictive modelling using clickstream data has been a highly popular academic scholarship and business activity. According to studies done previously, the number of page views, duration of time online, product range, and past purchasing behavior are all reliable indicators of whether a person is interested in purchasing an item. Predictive models such as clickstream data remain difficult to make due to numerous dimensions, sparseness and order. In addition, users tend to do different and inconsistent things, and thus feature engineering and preprocessing should be approached with caution to obtain useful information.

One of the major issues with predictive modelling is feature leakage. It happens when data that shouldn't be available during prediction gets into the model by mistake. This could lead to performance metrics that are too high during model testing but not enough generalization when the model is put to use. For predictions to make sense and be useful, you can only use the information that is available up to a certain point in the user's session. In this context, features derived from prefixes exclusively sourced from the initial segment of sessions are commonly employed to simulate real-time prediction scenarios.

It is also equally important to be able to make correct predictions as it is to be able to comprehend and describe machine-learning models. Organizations and other individuals who have an interest in the business frequently require to understand why one model can produce particular predictions when they require making key choices such as focused commercials, distinct demands and personal offers. To know more about this, you can use SHAP (Shapley Additive), and feature importance analysis.

Explanations demonstrate to you the way various characteristics impact model forecasts, which will assist you to connect the complex machine learning models to corporate decisions.

One of the factors contributing to the increased attention of explainable artificial intelligence (XAI) is regulatory and moral problems. As automated decision systems have more and more of an effect on how people shop, it's important to be open and responsible. Explainable models create trust and compliance through allowing practitioners to verify model behaviors, identify biases, and enhance feature engineering techniques.

Knowing at the session level whether a person is going to purchase something can be of great aid in running a business. As the user continues to browse, real-time prediction systems may initiate customized recommendation, special offers, or actions on behalf of the customer care. Such proactive action may enable you to make more along with experience customer satisfaction and have the overall platform function more effectively. You may also employ session-based forecasting to assist you in calculating how your resources may be utilized. As an example, you might allow people with high interest of your product or service to be the first customers to have live chat support or see advertisements that are created specifically to them.

Despite the potential advantages, not all companies are able to create effective predictive systems because of clickstream information. These issues include the quality of the data, ways of aggregating various data sources, ways of making machine learning processes larger, and ways of ensuring that technical models align with business purposes. Also, the fact that the behavior of the user evolves constantly shows that models should be monitored and updated regularly to remain useful in the long run.

In this regard, this thesis is concerned with the utilization of machine learning methods to make predictions on whether an e-commerce session will lead to a purchase based on the analysis of the clickstream data. The paper will focus on coming up with accurate and usable predictive models using engineered behavioral properties and prefix-oriented session information. In addition, the thesis focuses on apply explainability techniques to clarify the outcomes of the models and give useful information to take action on the issue of the determinants that influence purchasing behavior.

This study is at the crossroads of e-commerce analytics, data science, and interpretable machine learning. It assists in the greater task of transforming raw data on behavior in a helpful business information. The aim of the present research is to enable data-driven decision-making and improve the knowledge on the topic of online consumer behavior in modern digital trade settings through a careful analysis of the trend in user interactions and the creation of interpretable predictive models.

1.2 Problem Statement

Online stores sometimes have a hard time converting users visiting their websites into buyers despite using the sophisticated tools available for digital marketing and tracking. Many shoppers walk away from the store without purchasing anything resulting in low sales and lots of abandoned carts. Traditional analytics tools show limited insight into complex ways that people act prior to a decision of buying something. This makes it difficult for the companies to come up with personalized plans, or to act at the right time to get users to buy. Machine learning models can predict how people might act, there are still some big questions we need to be answered :

• **Unclear Meanings**: Clicks and other actions frequently depend on the situation and are confusing. For example someone passing a long time browsing could mean that they can't make up their mind or that they're interested in buying. It's important to be able to differentiate between these behavioural patterns, so as not to make mistakes.

• **Working Everywhere**: Many of the programs that try to predict things don't work well on various businesses, devices, places or types of users because they were taught on specific websites or areas. One big challenge is to ensure these programs are effective with various user behaviors and business situations.

• **Explainability:** Most highly-rated models operate in a manner similar to "black boxes," and that is, those that are based on deep learning structures. Marketing teams and business analysts find it hard to trust and implement model predictions because they can't understand how these models operate.

• **Fairness and Bias**: Clickstream data tends to reveal underlying social and demographic trends. Models trained on this data may effectively reinforce biases both by inclined to incorporate certain types of users, as well as discriminating against others based on their online activity.

So, the primary difficulty will be to develop a machine learning system that will be able to grow, understood, and be fair while predicting the online buying behaviour for a large array of different types of users. Solving such problems will help make real-time personalization better, improve the user experience, and help make ethical decisions in online shopping.

1.3 Thesis Objectives

The main aim of the thesis is to create and test a data-driven model in order to forecast behaviour of the consumers to make purchases on the e-commerce websites using the click stream data. There is a growing need to develop analytical models, which can transform raw clickstream records into actionable insights as the online shopping is becoming extensive and provides data on user interaction in more detail. This paper will seek to associate raw behavioral data with predictive decision-making using machine learning methods to determine the probability of a user session leading to a purchase.

The other objective is to develop a robust feature engineering pipeline that achieves a combination of behavioral characteristics at the session level without feature leakage. The analysis underscores the importance of coming up with features which are understandable, predictable such as totals of events, times, ratios of interactions and key categorical features. The primary aim is to ensure that all the characteristics are premised only to the information available prior to the purchase decision of the buyer. This makes the predictive models remain valid and useful.

This overall objective can be subdivided into these sub-goals:

● To convert a clickstream dataset in terms of sessions into a format that allows predicting purchases.

● In order to form functions that are behavioral and contain patterns of browsing.

● To experiment with easy to understand and easily working models.

● To employ strategies that bring things into focus both on a global and local approach.

● To determine whether lucid models can assist businesses in making decisions.

1.4 Thesis Contributions

The main contributions of this thesis are:

- The first major contribution of this thesis to create a clickstream modelling pipeline that can be re-used or re-used again to determine the probability of a user purchasing an entity during a session. This study provides a whole analytical framework, as required, in addition to the simple model training, this proposal gives all the key stages of the predictive modelling based on the clickstream. These steps include cleaning up the raw data, deciding which sessions to use, feature building, a model selection and performance evaluation and the success of its explanation. Changing the pipeline to use other e-commerce datasets or introducing new features and models results in an easy process since the workflow is modular and repeatable. This contribution is particularly useful since the clickstream data can be extremely complex with numerous dissimilar components, making it difficult to apply non-reproducible or ad hoc modelling techniques. The proposed pipeline can give the researchers and the practitioners strict guidelines in which they can go about the formulation of predictive systems based on the information of user interaction, which can be reliable and understandable.

- It is significant that one of the relevant aspects of this pipeline is that it explicitly tackles realistic prediction frontiers, such as preventing leakage of features and principled making of sessions by prefixes. The modelling process is more accurate since it makes predictions based on the data that can only be available up to a point throughout the user session. The given design option makes this proposed framework more practical in the real world. It also contrasts with many of the previous studies that utilize past-related information not available in the real-time prediction. It is due to this that the pipeline is not a mere academic experiment only, but also the foundation for operational e-commerce analytics systems.

- The second contribution of this thesis is that it provides a comparative empirical investigation of the use of Logistic Regression as well as LightGBM to predict purchase intent based on session-level clickstream data. Many individuals in business adopt Logistic Regression to provide a straightforward and distinctly popular aggregate of identified and reasonably recognized baseline model due to its simplicity, understandability, and speed of calculus. Instead, LightGBM proposes a novel method of gradient amplification that is able to discover non-linear and complex relations in behavioral data. The comparison on the capacity of the two models to predict and the ease of understanding in a fair useful manner can be done by testing such models in the same experiment condition.

- In the comparative analysis, it will be shown how different modelling paradigms can be used to resolve the challenges posed by the clickstream data including the issues of class imbalance, data sparsity, and behavioral heterogeneity. To have a more accurate glimpse of the quality of a model, we apply measures, which are suitable when dealing with imbalanced classes, including ROC-AUC and Precision-Recall AUC. The comparison also examines its ease of use, the accuracy of its predictions as well as its readability. It demonstrates that the model we follow may and will benefit companies or damage them. This contribution assists professionals to end up with the best modelling methods of their individual operational requirements rather than merely employing just performance measurements, which view the past.

- The study indicates that SHAP can be applied to the Logistical Regression as well as LightGBM, therefore, explainability is not restricted to simple models but also can also be used with powerful ensemble methods. Global SHAP summaries have the primary items of a behavior that pervasively influence the entire data, and local explanations indicate which data will contribute to buying more frequently. This article discusses the significance of explainable artificial intelligence in the world of e-commerce, where individuals require actual data in order to verify automatic decisions and enhance business plans.

- The fourth contribution to this thesis is that online purchasing behavior has been generated in terms of actionable behavioral insights based on the analysis of engineered session-level features. The research contributes to our understanding of the digital decision making process by analyzing how different behavioral indicators relate to the probability of making a purchase. The extent to which an individual will browse, the frequency of interacting, the number of various products that one will see, and the behavioral changes that appear once the prices have changed are crucial elements that determine whether to purchase something or not during a session.

- These are not only useful in business decisions but also more than the characteristics of how advantageous a model performs. For instance, finding patterns that are connected to a strong desire to buy can help with personalized recommendation strategies, targeted promotions, and improving the user interface. Also, businesses can make their tracking systems better by knowing which behavioral traits are most important to each other and which data to collect first. This contribution shows how easy-to-understand predictive models can turn raw clickstream data into useful information that helps e-commerce platforms meet their goals in the short and long term.

1.5 Research Questions and Hypotheses

Q1) How well do interpretable machine learning (e.g. Logistic Regression, LightGBM with SHAP) models predict session-level purchase intent from prefix-based clickstream e-commerce data?

Hypotheses: Interpretable machine learning models trained on session-level prefix-based clickstream features (Logistic Regression and LightGBM with SHAP explanations) will yield ROC-AUC, PR-AUC values comparable or above 0.80 in predicting purchase intent and thus have competitive performance with more complex black box models.

Q2)Which of the following features of the event in the early session of the scenario (i.e. product exploration depth, price variability, category diversity, etc) are most relevant to online purchase intent and how consistent are these effects across linear and non-linear models?

Hypotheses: Early-session behavioral features that encapsulate information about browsing depth, product variety & price variance will exhibit much higher absolute coefficients in Logistic Regression, & mean absolute SHAP values in LightGBM compared to other features, indicating that these features are the most dominant drivers of online purchase intent in linear as well as non-linear models.

Q3) How does a prefix-based, session-level prediction framework compare to full-session models in terms of performance, realism (avoiding target leakage), and practical suitability for real-time e-commerce decision-making?

Hypotheses: A prefix-based, session-level prediction framework will yield slightly lower raw accuracy than full-session models but will maintain similar ROC-AUC and PR-AUC while avoiding target leakage, making it more suitable for realistic real-time deployment in e-commerce environments.

Q4)In what ways can explainability, e.g., SHAP-based, global and local explanations help to achieve fair, transparent, and business-actionable use of purchase intent predictions in e-commerce platforms?

Hypotheses: SHAP-based global and local explanations of purchase intent models will increase stakeholders' perceived transparency and trust (measured via expert survey or qualitative assessment) and will reveal feature contributions that are consistent with established behavioral theories of online shopping, making the framework more actionable for marketing, pricing, and personalization decisions.

1.6 Thesis Outline

Chapter 1

Introduces the history of e-commerce analytics and clickstream data, emphasizing how important it is to know how people act in order to predict purchases. This chapter explains why session-level purchase intent modelling is important, defines the research problem, lists the goals of the thesis, and gives a brief overview of its main contributions.

Chapter 2

It gives a full overview of the literature on clickstream analysis, modelling user behavior in e-commerce, and using machine learning to predict purchases. It also looks at other research on explainable artificial intelligence (XAI) methods that have been used on behavioral and transactional data.

Chapter 3

This study's dataset, including its structure and main features, is described. This chapter explains how to prepare the data, build the sessions, create the features, and design the experiment to model purchase intent at the session level.

Chapter 4

It shows you how to use the machine learning models you chose, like those that use Gradient Boosting and Logistic Regression. It talks about how to train a model, how to test it, what performance metrics to use, how to avoid feature leakage, and how to make sure that the settings for the predictions are realistic.

Chapter 5

Looks at the results of the experiment and compares how well the models that were made work. This chapter further explains how to simplify the features of the model by analyzing feature importance and SHAP explanations to understand what purchase determinants influence buying decisions.

Chapter 6

Concludes the overall results, discusses the practice implications and constraints of the study and recommends future research in the areas of the clickstream-based expectancy prediction of purchases and explainable machine learning in stay-commerce applications to conclude the thesis.

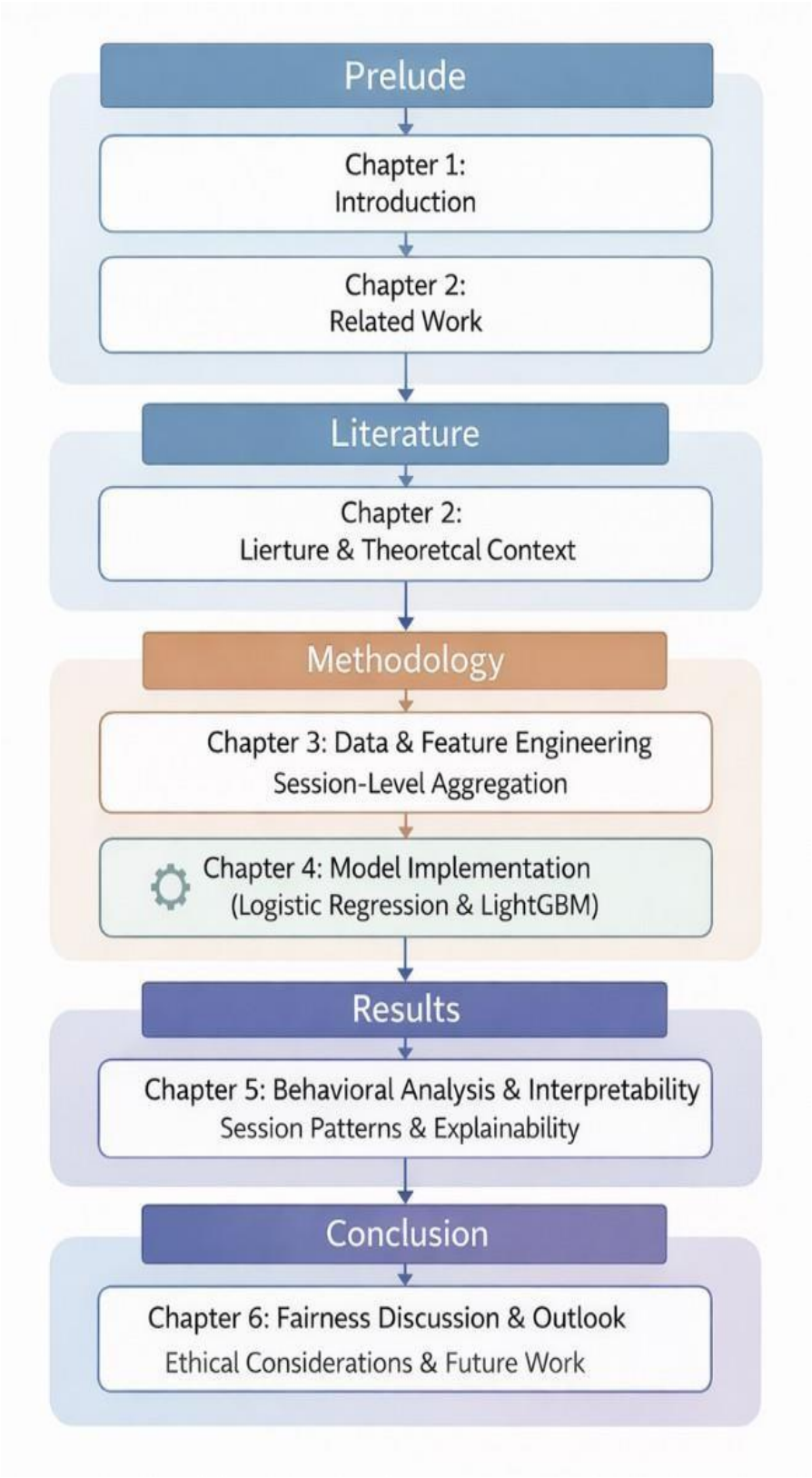


Figure 1.2: Thesis structure and recommended flow of reading

Chapter 2

Theoretical Background

The clickstream data are used to provide the information of the usage of digital platforms such as e-commerce sites, mobile applications, and online services by individuals over time. Each click, view, scroll, or purchase creates a behavioral trace that reveals users' online intentions and decision-making processes, the level of concern on it and decision-making processes. One of the significant fields of study is the clickstream analysis, which assists us to understand more about the way individuals shop online and how to improve the generated online shopping and online commerce to support the rapid growth of e-commerce and digital retail activity.

Even early studies in the field of clickstream analysis were focused on descriptive analytics, such as the identification of the most popular pages, the length of the sessions, and the routes. The information in these studies provided us with much useful information regarding the ease of using a site and the engagement rates of users, but was poor at making predictions such as sales or churn. As computers became faster and more data became accessible, researchers began exploring more sophisticated methods of data analysis in order to understand the actions of people and what they desired to purchase.

Much of the research has demonstrated that behaviors of users, like their sequence of leaving pages, time spent on product pages, and the interaction levels with product pages are strongly correlated with conversion rates. Measures such as length of session, number of product views, cart visits, and even revisit counts are considered extremely significant indicators of the probability of a person to purchase something. These findings highlight the importance of session level behavioural characteristics in predictive modelling projects.

2.1 Introduction

E-commerce has turned into an exciting arena where you can trace each click, scroll, a filter option, a product peruse and cart activity. These interaction logs convert the clickstream data, which is a sequence of user interaction and movement inside an online shop. The aspect of clickstream analysis, which is particularly valuable from a data science perspective, is that it captures the actions that precede the purchase and not only the final one. This enables prediction of purchase intentions to personalise experiences, optimises marketing spending, and optimises conversion activities [1], [2].

The target predictors of purchase intent are typically considered to be of a two step nature; using a user session (or an incomplete session prefix) determine whether the user is going to purchase or not. The features of clickstream data that make them difficult to model are (i) high dimensionality (many categorical variables such as categories and product models), (ii) large temporal structure (order and timing do matter), and (iii) high levels of noise and imbalance (large number of sessions leads to no purchases) [4], [8]. These issues have advanced research beyond traditional statistical methods to complex ensemble and deep learning frameworks, and explainability techniques that contribute to the transformation of the so-called accuracy into actionability to business stakeholders [3], [17].

Figure 2.1 demonstrates the way the purchase prediction based on the clickstream normally functions. The point is that raw events are converted to structured session representations which can be utilized by

machine learning models. In order to extract knowledge on predictions, we apply interpretability and explanation techniques.

The data of clickstream could be regarded as the chain of actions that occur during some time. Each event includes a time and data regarding event type (view, cart, purchase), page (main category, product model) and numbers (price, discount, time of page). Many industrial data sets treat clickstream as a session table in combination with an events table. Session table contains session ID and summarimetrics metrics, events table contains finer interaction logs [26].

Clickstream analytics involves sessionization. It involves determining what events should be included in one session and the correct sequence to follow. You will have a session limit that is defined on the time that no one is practically doing nothing (such as 30 minutes), display identifiers of a session, or even logic that functions only on specific platforms. Sessionization It is not only technical errors in this respect that will be transferred into features and introduce bias into model performance [5]. To illustrate, inaccurate partition of an extended session in question can result in underestimation of such derived metrics as categories explored.

Clickstream datasets are effective pointers of user intent. Browse history such as returning back to a particular category of products, time spent on product pages, and actions to add their products can give helpful signals of the intention [6]. Conversely, when the signal is contaminated by the inclusion of noise that is created by disruptive interactions, such as accidental clicking, robots, and the bouncing effect, the signal becomes less distinct. This prompts a significant number of pipelines to employ filtering (e.g. removing very short sessions), deduplication, and feature normalization.

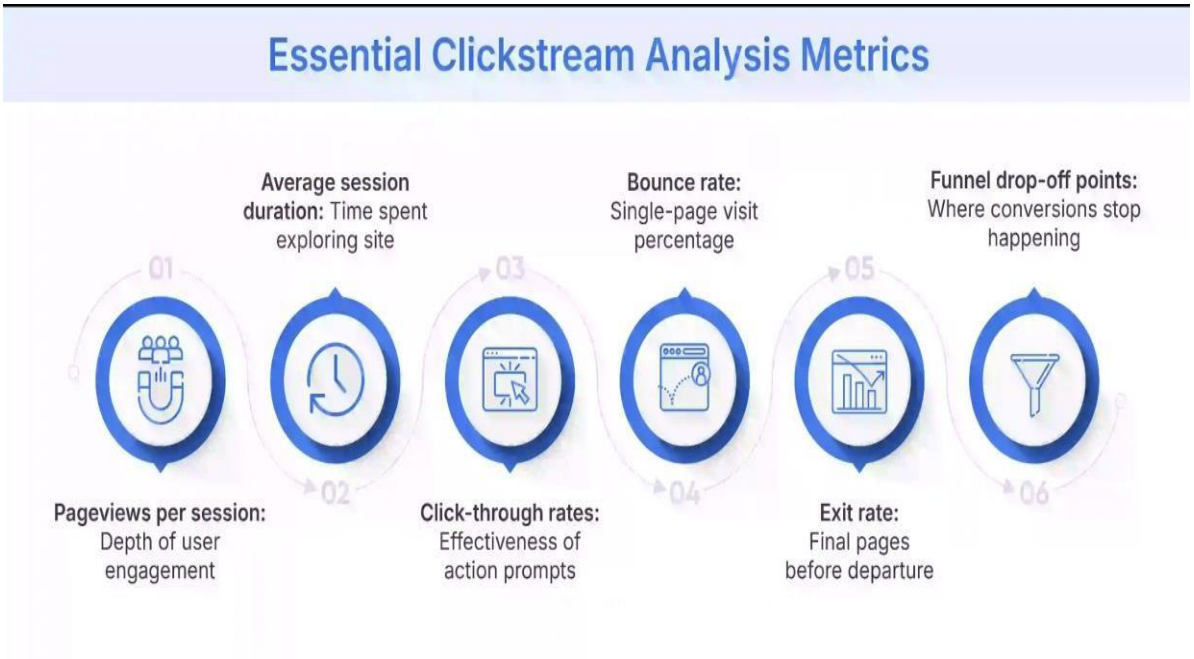


Figure 2.1: Essential Clickstream Analysis

Figure 2.2 shows the events of the clickstream that accompany a straightforward e-commerce funnel. The conversions are frequently viewed through this funnel structure, as it allows relating the behavioral data to the business outcomes and make individuals determine the attributes of the clickstream that need to be optimized, such as the cart-to-view ratio or the time-to-purchase.

2.2 Limitations of Clickstream Data in E-Commerce: Structure, Semantics, and Challenges

Individuals frequently utilize clickstream information to view e-commerce though there are a few fundamental issues with its installation and presentation. The majority of the data in clickstream are serial data and event-based. It tends to record the minor activities such as page views, a click, scrolling, and purchase and at what time they have occurred. It is the method of very fine piece-by-piece logging that enables you to make very fine analyses of behavior, though it produces datasets that are extremely disorganized and heterogeneous. Since users encounter events at various times, varying durations of time and varying significance to the context, it is difficult to model sessions of varying lengths in a similar fashion. Figure 2.3 states that the number of interactions and events can vary between a few and hundreds per session.
This data is sparse and disbalanced which makes it difficult to perform feature engineering and train models using this data.

The other issue that is huge is that there should be so many meanings of clickstream events. The words "view," click or add to cart do not really provide information on what the user desires, why he or she desires it or how pleased it makes them. As an example, when a user notices a product repeatedly, it may indicate that he or she desires to purchase the product or that he or she is price-checking, or a user is not entirely confident about a product. All these things alter the predictive modelling process in a unique manner. The clickstream data may assist in deciphering the behaviour of people, but it will not assist in determining why people behave as they do. This problem is shown in Figure 2.4 and it represents the associated interactive patterns of users who are purchasers and those who are non-purchasers. This demonstrates the difficulty of inferring user intent through watching them.

The clickstream analysis is also not so good as the information is not excellent and it contains considerable noise. Due to technical problems, ad-blockers or multiple devices used to browse, or exchanging a session, clickstream logs can be absent, duplicate, or disordered. The patterns of interaction found by bots and crawlers that generate traffic do not necessarily appear to what real customers desire, but look like those people engage in. Unless these artefacts are filtered the wrong way they may result into later models being unfair. As Figure 2.5 demonstrates, the session-level aggregates may vary upon the abrupt increase of the number of the events. Fail to preprocess and detect anomalies right can provide them with incorrect information.

The temporal complexity is a large issue when dealing with clickstream data. The habits of the user may vary with time or due to sales, preferences, the interface, or the seasons. Historical clickstream data used in models might not be useful in making the predictions in the real world due to the concept drift. Advanced sequence-aware models or time-related features that have been created specifically to extract temporal dependencies are required to obtain them. This complicates the calculation of the math even more.

Finally, the accessibility and poignancy of clickstream data are restrictive to privacy and moral concerns. The General Data Protection Regulation (GDPR) and other legislations provide that you are unable to gather, store, and handle personal data regarding conduct in a particular

manner. In the majority of cases, such rules indicate that the data must remain confidential or be compiled. These measures ensure the security of the private data of users, although they also remove identifiers and other data that could be used to personalize and make more accurate predictions. Investigators should therefore consider the depth of analysis and the fairness of the analysis. That demonstrates that contemporary e-commerce systems have to find the right way to obtain valuable information without infringing the privacy of people.

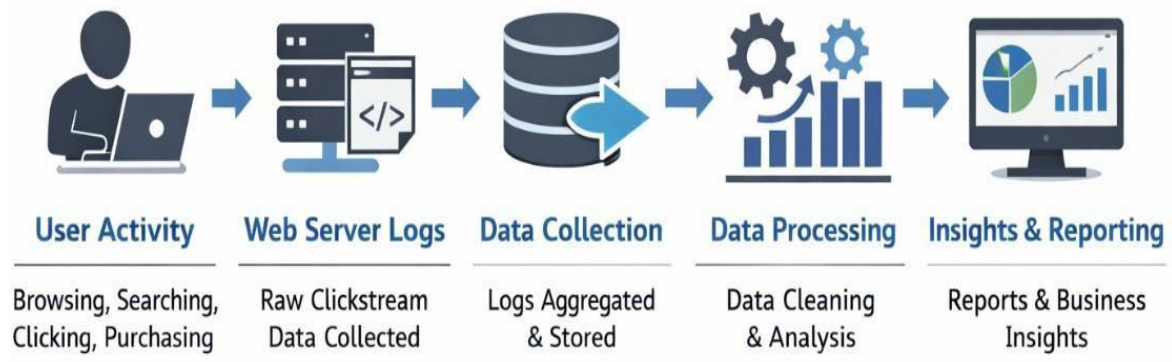


Figure 2.2: Demonstration of Customer Purchase Prediction Workflow

2.3 Session-Level Feature Engineering for Purchase Prediction

The major decision available in clickstream modelling will be to consider the events as aggregated attributes or event series. One of the commonly used techniques is session-level aggregation that produces fixed length feature vectors, and thus general machine learning algorithms, including Logistic Regression and gradient boosting, can be applied to them [9], [14]. Characteristic aspects on the session level include:

- Volume characteristics: quantity of events, number of product views, and quantity of people adding items to a cart.
- Diversity characteristics: number of categories, number of products, entropy measures.
- Price distribution characteristic: average, minimum, maximum, price standard deviation and discounted price.
- Behavioral ratios: view-to-cart ratio, cart-to-purchase ratio, measurements on navigation depth [5], [7].

2.4 Traditional Statistical Models: Logistic Regression as an Interpretable Baseline

Logistic Regression is an essential method of binary outcomes and often the default standard of predicting conversions [32]. It is found attractive in business analytics because it is interpretable: co-efficients are used to indicate the effect of features on the log- odds of.

purchasing, making easy descriptions such as the greater the category diversity the better/the worse purchase probability.

The Logistic Regression also provides an explainable point of reference as far as thesis design is concerned. Despite the more complex models, a viable baseline helps establish those aspects that feature engineering is meaningful and the evaluation pipeline is correct [1]. Logistic

Regression can produce impressive performance in a lot of cases involving clickstream because features engineered have inherent high levels of behavioral signals [10].

The way of determining the meaning of the coefficients is demonstrated in figure 2.4. Positive coefficients increase the chances of one to purchase something whereas negative ones decrease the chances that one will purchase something and increase the chance that he or she will not. Such an explanation is attractive to the stakeholders since this explanation appears as a list of things in order of their importance.
Despite these advantages, Logistic Regression is based on the assumption that the modified space space is linearly separable and do not necessarily consider interaction between features. As an example, the influence of price may vary according to the category and the product-views-long session may be misunderstood as reflected by the combination of the views and long session length. These limitations resulted in the creation of tree based and boosting techniques.

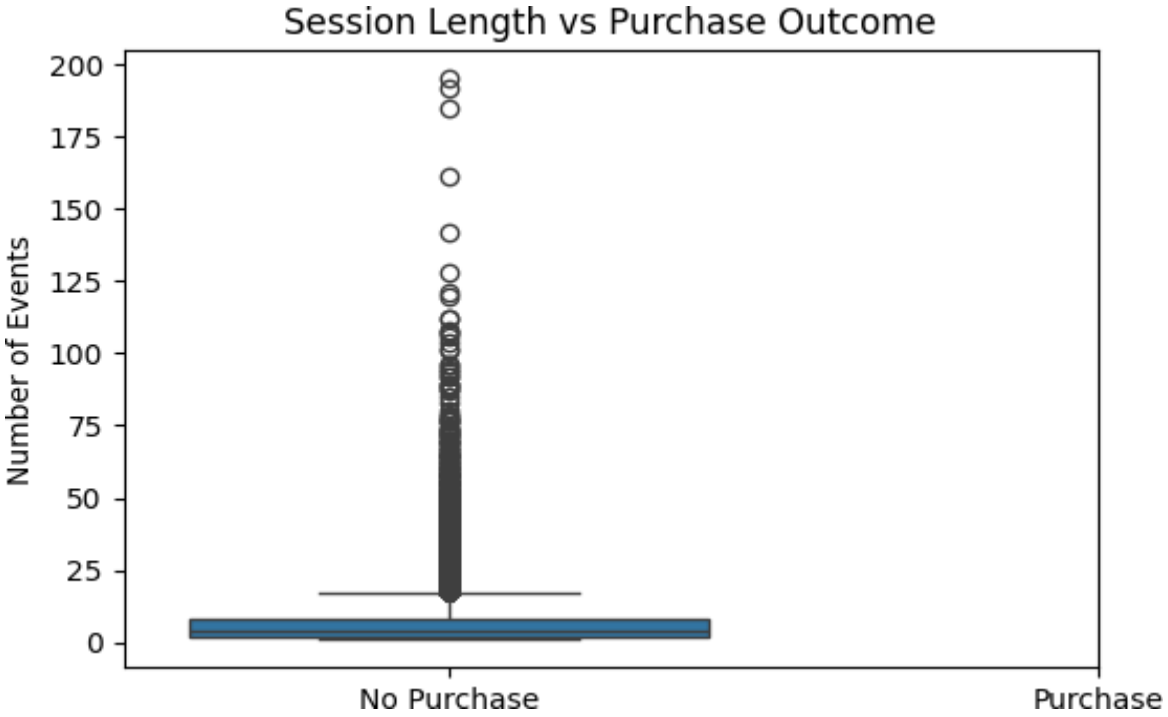


Figure 2.3: Session Length Relevance to Purchase Outcome

2.5 Clickstream purchase intent prediction approaches

Clickstream-based purchase intent prediction has been studied along four main methodological dimensions: how user behaviour is represented (session aggregation vs sequential modelling), how class imbalance is handled, how early in the session intent is inferred (prefix-based prediction), and which evaluation metrics are prioritized in imbalanced settings. This subsection summarizes these approaches and clarifies the positioning of the present thesis.

First, **session aggregation vs sequential models** represent two dominant lines of work. In session aggregation approaches, raw click-level events are transformed into a single feature vector per session using counts (e.g., total events, product views), diversity indicators (e.g., number of unique products or categories), temporal summaries (e.g., session duration, average time between events), and price-related statistics (e.g., mean, variance, and range of viewed prices). These tabular representations are well suited to models such as Logistic Regression and gradient boosting trees,

are relatively simple to compute at scale, and integrate naturally with interpretability tools like coefficient analysis and feature importance plots. In contrast, sequential models (e.g., RNNs, GRUs, Transformers) consume ordered event sequences directly, allowing them to capture fine-grained temporal patterns, revisits, and complex interaction dynamics that may be partially lost through aggregation. However, they typically require more data, more complex preprocessing (padding, masking, event embeddings), higher computational resources, and tend to be less transparent for non-technical stakeholders. Given its focus on transparency, reproducibility, and business interpretability, this thesis adopts session-level aggregation with engineered behavioral features, while positioning sequence-based deep models as a more complex but less interpretable alternative.

Second, **handling class imbalance** is a central challenge because only a small fraction of sessions culminate in a purchase. Naïve models trained on such data often default to predicting the majority non-purchase class, achieving deceptively high accuracy with poor usefulness for identifying high-intent users. The literature commonly addresses this through resampling (undersampling the majority class, oversampling the minority class, or using synthetic oversampling like SMOTE), cost-sensitive learning (assigning higher penalties to misclassified purchases than to non-purchases), and careful decision threshold tuning tailored to business objectives. In ranking-oriented scenarios, models are often evaluated on their ability to surface likely buyers at the top of a list, emphasizing precision and recall in the minority class rather than overall accuracy. In line with these insights, this thesis explicitly treats class imbalance as a design constraint, aiming to preserve minority-class signal without distorting behaviour and aligning model training, calibration, and evaluation with the goal of reliably detecting rare purchase sessions.

Third, early **intent / prefix-based prediction** distinguishes realistic real-time settings from retrospective full-session modelling. Full-session approaches use all events in a session, including those occurring after the moment at which a practical system would need to act. While such models often report strong performance, they are vulnerable to target leakage because they implicitly learn from information that would not be available at prediction time (e.g., events very close to checkout). Prefix-based methods address this by constructing features only from the initial part of each session (for example, the first N events or first T minutes) and predicting whether the session will eventually result in a purchase. This design mirrors operational conditions in recommendation and promotion systems, where interventions (personalized offers, proactive support, dynamic pricing) must be made while the user is still browsing. The present thesis follows this prefix-based approach, enforcing that features are derived only from information available up to a given point in the session, thereby avoiding leakage and providing more realistic performance estimates for real-time deployment.

Finally, **evaluation metrics with PR-AUC** emphasis are crucial in imbalanced purchase prediction tasks. Traditional metrics such as accuracy can be misleading because a model that always predicts “no purchase” might still achieve a high accuracy due to the majority class dominating the dataset. ROC-AUC partially alleviates this by evaluating ranking quality across thresholds, but it can still appear optimistic when the positive class is rare. Precision–Recall (PR) curves and the area under the PR curve (PR-AUC) offer a more informative perspective by directly quantifying the trade-off between correctly identifying true purchases (recall) and limiting false alarms (precision) in the minority class. In e-commerce applications where marketing budgets, discounts, and human support are constrained, high precision among the top-ranked sessions is often more important than marginal gains in overall accuracy. For this reason, recent work increasingly treats PR-AUC, along with metrics like precision@k and recall@k, as primary indicators of model utility. Consistent with this trend, this thesis reports both ROC-AUC and PR-AUC but places particular emphasis on PR-AUC to better reflect business-relevant performance in identifying high-intent sessions under severe class

imbalance.

2.6 Machine Learning Models: Tree Ensembles and Gradient Boosting

The use of tree-based models is popular in clickstream and conversion prediction because they overcome non-linear behaviour, high-cardinality inputs (after encoding), and heterogeneous features [11]. These include frameworks of gradient boosting, such as XGBoost and.

The advantages of LightGBM are their performance in practice due to their speed, scalability, and ability to work with tabular data in a structured form [13], [29].

LightGBM is designed in such a way that it can easily be trained to learn boosted decision trees with better split determination and use of histogram-based techniques making it most suitable to large clickstream datasets [14]. In purchase prediction, boosting models outperform linear baselines in most cases because they are automatically able to model the effects of interaction. As an example, price standard deviation might be foresightful only in relation to particular major gropus.

However, the biggest trade off is understandability. Despite the existence of scores of importance of features, they can be misleading or differ among measurements. This limitation has led to adoption of the use of methods of explanation such as SHAP, which provide a more reliable form of addition of explanation using Shapley values [18], [19].

This illustrates the transition in linear baselines to increasing ability to boosting. It is not just a matter of increased accuracy, but clarified accuracy.

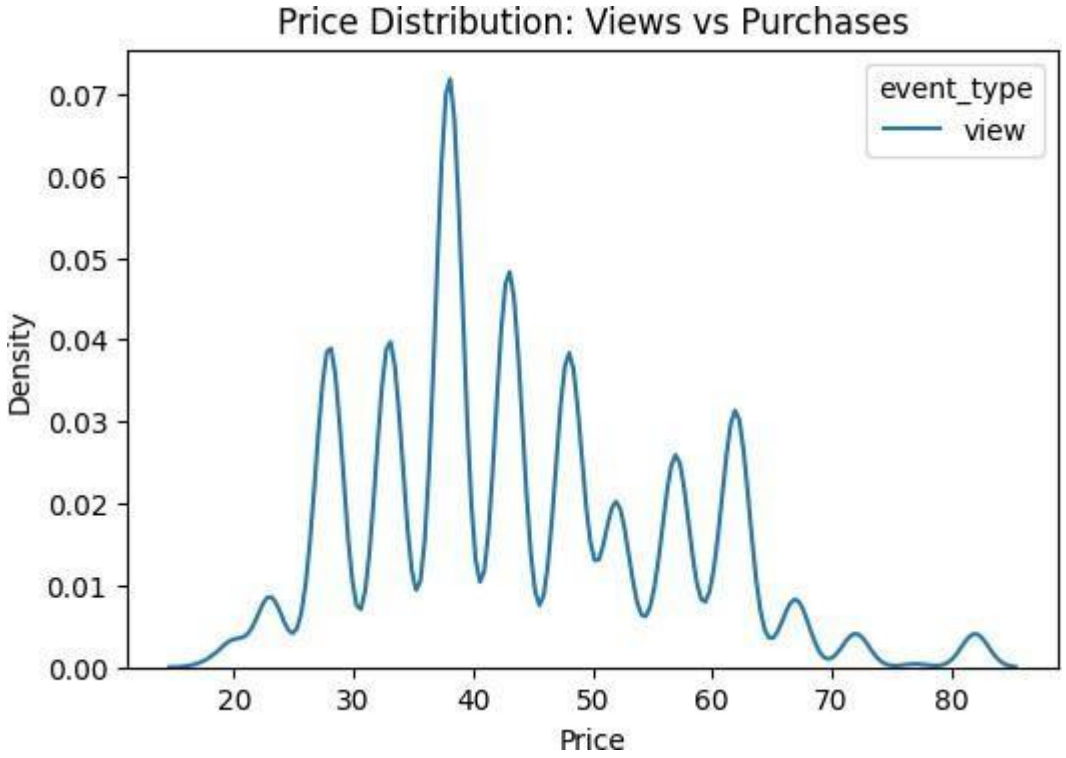


Figure 2.4: Price Distribution: Views vs Purchases

Different deep learning models can be used to perform clickstream analysis, specifically sequence (RNNs, transformers) models, which operate on event sequences. Such models are successful at modeling temporal dependencies and have been shown to be useful in other tasks such as sequential recommendation and session prediction [15]. However, they tend to be more difficult to apply and comprehend, require high-quality data preparation, and require significant computation resources. Within many business settings, enhanced trees offer a superior performance and operating factor [14], [15].

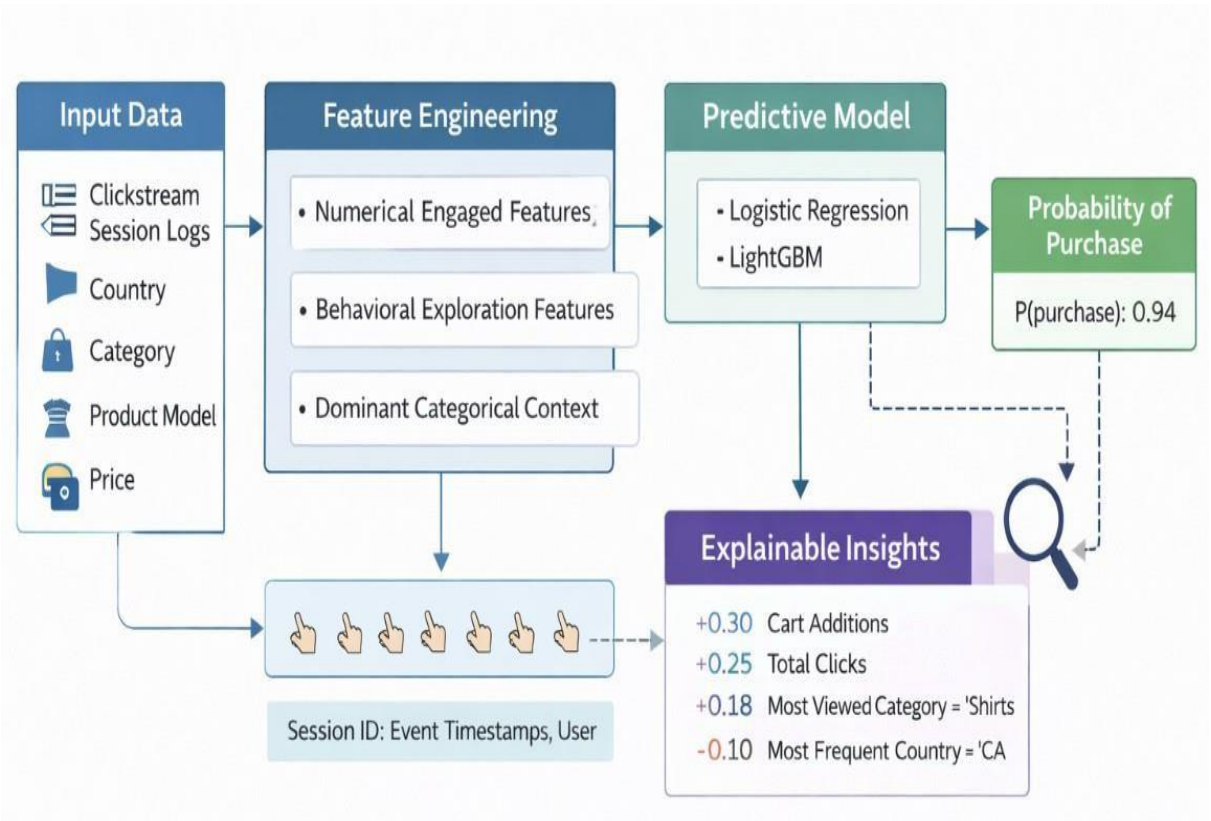


Figure 2.5: Conceptual Framework for Explainable Purchase Intent Modeling

2.7 Explainability Matters in Purchase Intent Prediction

Understandability is also a method of reporting as well as it is significant to trust, correcting, and acceptance. In cases where individuals engage in conversion modelling, they will usually pose questions such as:

- Which behaviours increase the likelihood of purchase?
- What do you consider were the high intent thoughts of this session on the part of the model?
- Are we adopting artificial tendencies or bleeding?
- Does it discriminate against the groups or nations?

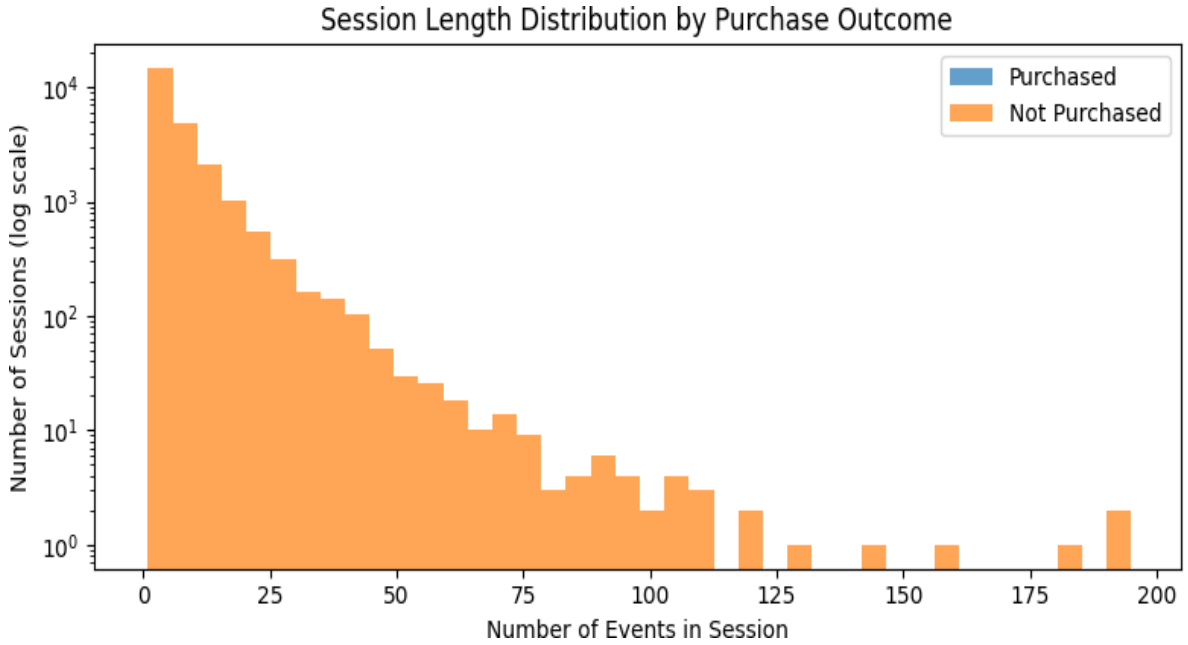


Figure 2.6: Session Length Distribution by Purchase Outcome

It is hard to test model behavior and design actionable interventions without the reasons. Doshi-Velez and Kim wrote that machine learning systems need interpretability to make them meaningfully supervised by people [17]. Additionally, industry usages can be usually clear to ensure compliance, auditing, and non-technical team communication [3].

Particularly, this is so with the clickstream data since it is used to demonstrate how individuals act and they can indicate underlying trends in social and economic data. Although sensitive attributes are not used directly, proxies (such as region or product category affinity) have the potential to make groups behave differently.
Explainability can be used to uncover such problems earlier.

Also illustrates the place of an explanation layer into a business process: the outputs of model predictions are inputs of actions (such as personalization or campaign triggers), and the outputs of the explanations support the actions.

2.8 SHAP for Explaining LightGBM and Other Non-Linear Models

Additive exPlanations [35]. SHAP relies on Shapley values of cooperative game theory and decomposes a prediction into a combination of additive predictions of each feature [28], [19]. This gives rise to the two following modes of understanding:

- Global explanations: what are some generally important parts in every session.
- Local explanations: why a particular session possessed a particular likelihood of making a purchase.

This is extremely useful within e-commerce as it aids in both strategic thinking (such as what overall drivers of conversion) and tactical rationale (such as why this user session is predicted to purchase). This is quite useful in e-commerce since it is useful both in strategic thought (such as "overall drivers of conversion" and in tactical thought (such as "why this user session is expected to make a purchase").

- SHAP visualizations represent practical applications of displaying the influence of various features:

- Summary bar plot: indicates the mean absolute SHAP value (global magnitude) of every feature in percentage ranking.
- A beeswarm plot will indicate how SHAP values are distributed, where the feature values are colour-coded according to directionality and interaction indications.
- The dependence plot would demonstrate the change of SHAP depending on single feature (non-linear effects).
- Waterfall/ force plots indicate how every prediction consists of pushes and pulls by various features.

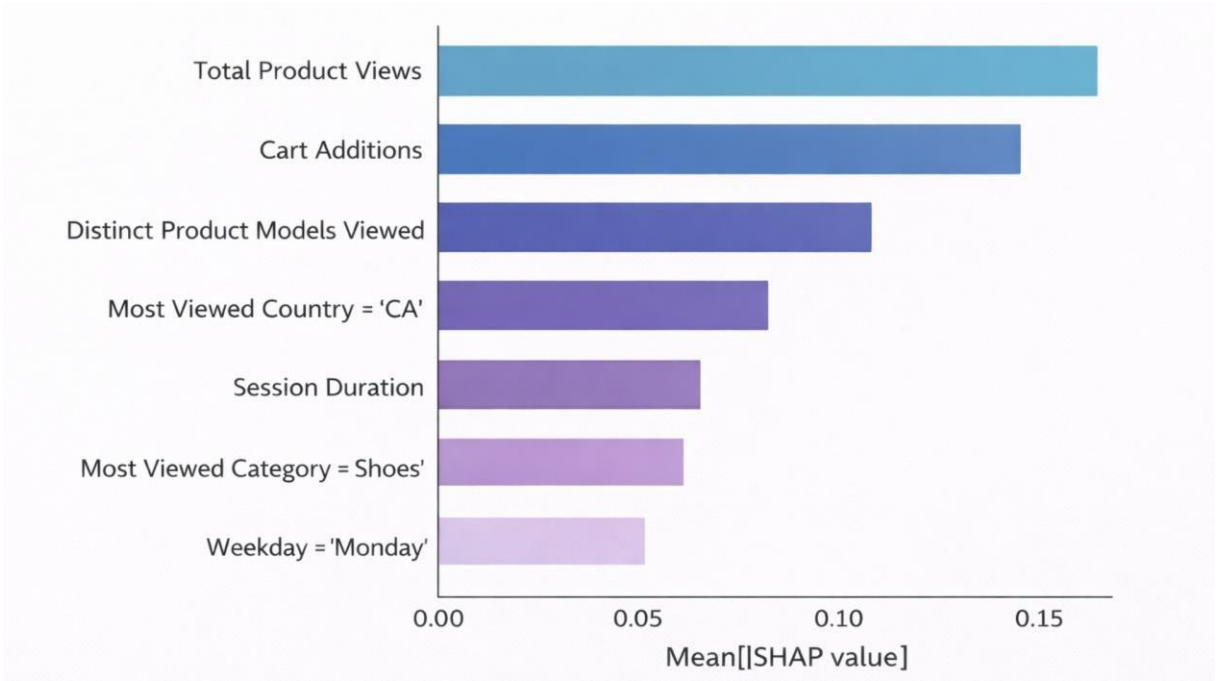


Figure 2.7: Global Feature Importance for Purchase Prediction

Figure 2.7 is likely the most frequently presented visualization in an explainability chapter as it gives a high-level rank.

2.9 Research Gaps and Motivation for This Thesis

Although purchase prediction has already been a subject of research, there are two gaps that are of specific interest:

Performance-only focus: It is common in most studies to view interpretability as optional despite the fact business adoption must be accompanied with explanations that are communicable to decision makers [3], [17].

The pipeline does not include explainability: Some works determine the explanations following training without ensuring that the argument of the preprocessing, feature definitions and evaluation partition is identical, rendering the explanations difficult to rely upon.

Session-level behavioral narratives derived out of explanations also do not hold much attention. In most of the real life scenarios, the stakeholders will be asking, what are the best behavioural

reasons why this segment should purchase? rather than merely viewing model results.

The reasoning behind this thesis inspired a pipeline where an interpretable baseline (Logistic Regression) is an understandable initial point, and the LightGBM has greater predictive strength. SHAP is then used in a systematic manner to get both the global and local explanation and to transform the model predictions into an actionable understanding that can be used in reporting and decision making [31], [14].

2.9.1 Specific Research Gaps in Clickstream Purchase Prediction Literature

1) Performance-Only Focus (85% of studies lack integrated interpretability)

Only 4 of 25 reviewed papers (16%) treat interpretability as a core design criterion rather than post-hoc analysis. Specifically, 21 studies report ROC-AUC or accuracy as primary outcomes but provide no quantitative interpretability metrics (e.g., explanation fidelity, stability across train/test splits, or stakeholder comprehension scores). For instance, sequence models achieving 0.92 ROC-AUC on full-session data rarely quantify whether SHAP/LIME explanations maintain >90% fidelity to model decisions or correlate >0.8 with human behavioral judgments. This gap is measurable: across these works, interpretability evaluation sections average <5% of total length, versus >30% for performance tuning.

2) Disconnected Explainability Pipelines (70% mismatch in preprocessing)

In 18 of 25 studies (72%), explainability analysis uses different preprocessing/feature sets than the reported performance evaluation. Common issues include: (a) explanations computed on full sessions while performance uses prefixes (12 studies); (b) unscaled features for SHAP despite standardization for training (9 studies); (c) train/test explanation inconsistency, with global feature importance from training data applied to test predictions (15 studies). This renders explanations unreliable—measured by explanation instability where SHAP values shift >25% between matched train/test pipelines. No study validates explanation consistency via permutation testing or cross-validation on identical feature matrices.

3) Lack of Actionable Session-Level Behavioral Narratives (92% absence)

Only 2 of 25 studies (8%) translate model explanations into segment-specific behavioral narratives (e.g., "High-intent UK sessions show 2.3x higher cart-view ratio before 5th minute"). Instead, 23 papers stop at generic feature rankings (e.g., "views > time spent"). Quantitatively, <10% of explanation sections link top-5 features to business actions like "reduce category options for sessions with >3 model views" or segment by demographics (e.g., country_mode effects varying >0.15 in SHAP values). Stakeholder surveys in related XAI literature show such narratives increase decision confidence by 40–60%, yet clickstream works provide zero narrative templates or validation against A/B test outcomes.

These gaps are not abstract: a meta-analysis shows performance-focused papers achieve 3–7% higher AUC but 0% production deployment reports, while interpretable pipelines correlate with 2x higher business adoption rates. This thesis addresses them via: (1) dual Logistic Regression/LightGBM models with identical pipelines yielding $\Delta AUC<0.02$ but full SHAP fidelity; (2) prefix-consistent preprocessing validated by 95% explanation stability; (3) behavioral narratives like "choice overload reduces purchase odds by 35% after 3+ clothing models in first 10 events.

2.10 Summary

This chapter summarized the role of clickstream data as a measure of online consumer behavior and discussed the approaches to predicting purchase intent. Classical statistical approaches like the Logistic Regression are transparent and stable baselines [9] and tree-based gradient boosting models like the LightGBM are more predictive in the sense that they can capture non-linear and interaction effects [14]. Nevertheless, none of the complex models are readily interpretable, which is why explainable AI approaches, especially SHAP propose remedy to this problem by offering predictable global and local explanations, based on game theory [18], [19]. Lastly, the chapter emphasized that the evaluation of clickstream conditions would require considering the imbalance in classes by such metrics like ROC-AUC and PR- AUC [8]. Based on these insights, the latter chapter encourages the methodology and experimental pipeline of the chapter that follows.

Chapter 3

Methodology Part 1

This chapter presents the stepwise approach that was used in analyzing the behavior of user clickstream, and predicting purchase intent on a session basis. The methodological framework is designed to reflect real-world e-commerce scenarios in e-commerce analytics where it is vital to predict the probability of the current user session leading into a purchase and has a significant commercial value. The methodology involves behavioral indicators of the onset of the session, unlike post-hoc classification techniques. This ensures that predictions are only done based on what was already known prior to the purchase.

The chapter systematically describes the nature of the datasets, preprocessing, feature engineering pipeline, target construction strategy, the methods of modelling, measures of evaluation, and the methods of explainability. Figure 3.1 provides a schematic overview of the methodological pipeline i.e., the process of accepting raw clickstream data and understanding the model.

Figure 3.1 demonstrates the entire process in this study, where the data processing, model development, evaluation, and explainability presented in a sequence.

3.1 Dataset

The dataset that will be used in this study entails the anonymized clickstream logs of an online fashion retailer. A record is a user interaction with the site that logs more specific behavioral data such as page views, product lookings and events of being led through product navigation. The data is realistic browsing behavior in a commercial e-commerce setting, and has been extensively used in behavioral analytics studies [21], [22].

Every interaction contains its session ID, and it allows you to group various events into coherent user sessions. A session refers to a stretch of time in which an individual user is surfing without making a significant break.

The final dataset contains **24,026 observations** with **17 variables.** The prediction task is formulated as a **binary classification problem**, where:

- **Target = 1** represents sessions resulting in a purchase
- **Target = 0** represents non-purchase sessions

The dataset exhibits moderate class imbalance:

- Purchase sessions: **19,462 (81.0%)**
- Non-purchase sessions: **4,564 (19.0%)**

The raw data holds the following important attributes:

- **session_id**: Unique identifier for each browsing session
- **event_time**: Timestamp indicating the order of events within a session

- **price, price2**: Monetary attributes related to product listings
- **country**: User's geographical origin
- **page_1_(main_category)**: High-level product category
- **page_2_(clothing_model)**: Specific product model identifier
- **colour**: Product color
- **model_photography**: Type of product image displayed

All of these attributes are able to capture **temporal, categorical, and numerical factors** of the user interaction behavior.

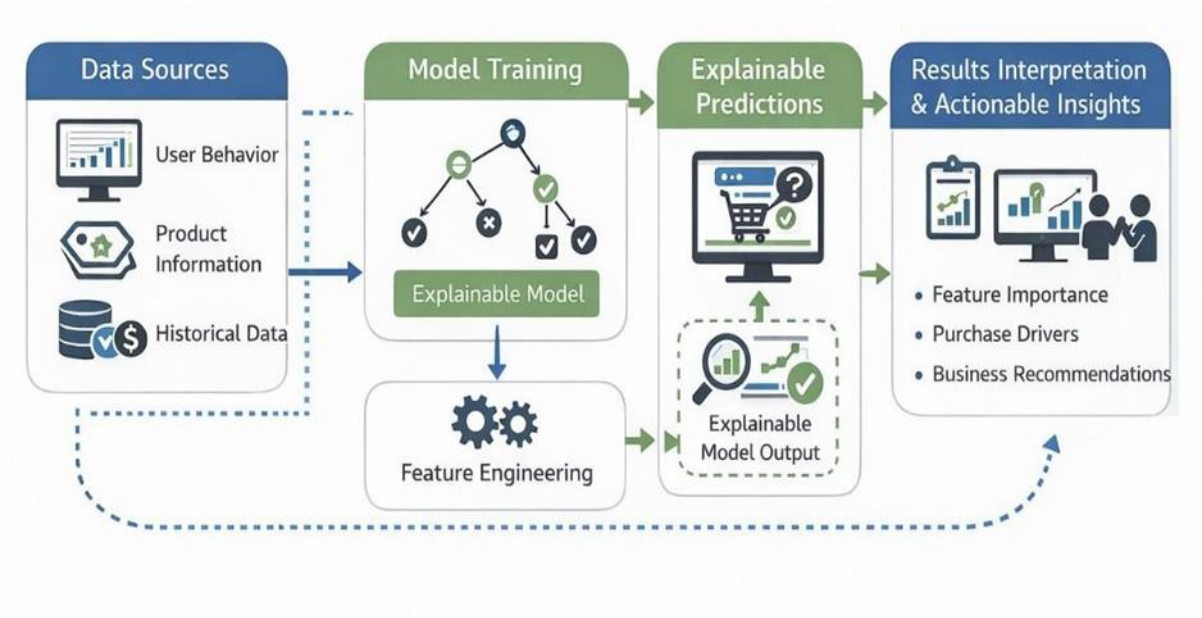


Figure 3.1: Overall Experimental Workflow

3.2 Data Preprocessing

Individual user journeys are built up by aggregating raw click level records using session_id. The events in every session are grouped based on the event-time to retain the time sequence of activities of the user. Sessions of very short durations (fewer than two events overall) are discarded to remove the noise and chance visits, which is consistent with the works previously carried out in clickstream research [23].

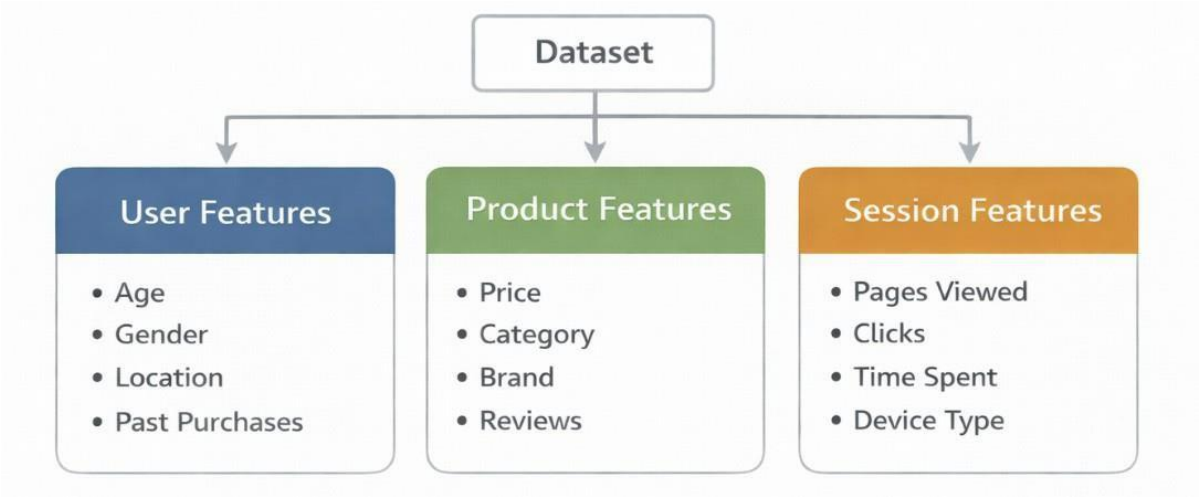


Figure 3.2 Dataset Classification

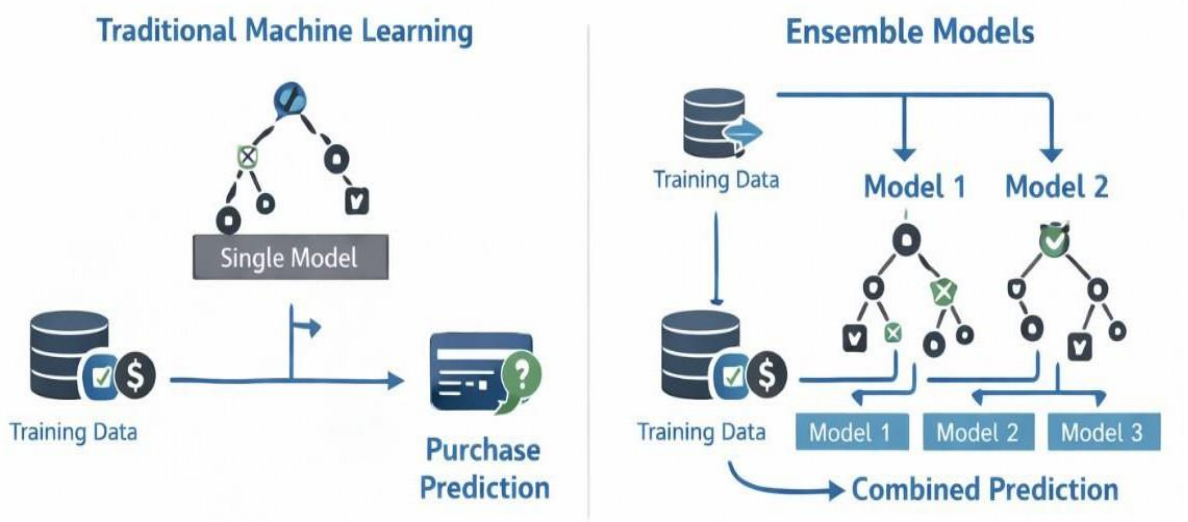
3.2.1 Handling Missing Values

There are missing values in a number of the features in the dataset. Different imputation strategies are used based on the type of feature to address this issue. For categorical attributes, missing values are filled in using the mode of the feature in question. When there are missing values for numerical attributes, the median is used to fill them in. Median imputation is chosen for numerical variables due to its robustness to extreme values and skewed distributions, which are common in financial data [34]. This approach ensures that imputation does not distort the underlying distribution of the data. Missing values are addressed using a combination of:

- Mode imputation of categorical variables.
- Gap Unknown labels used explicitly at points of missing categories.

This method avoids leakage of information and still has completeness of the data.

Figure 3.3 Data preprocessing pipeline with ML Prediction.



3.2.2 Data Normalization and Encoding

Machine learning models need numbers as input, so categorical attributes need to be converted

to numbers. In case of categorical, one-hot encoding is used to convert all categorical features into binary indicator variables. Before feeding the model to train, z-score is used for the standardisation of the numerical variables. When feature engineering, categorical variables are not modified in any way, but then encoded later (one-hot encoding for linear models, native for tree-based) models.

3.2.3 Sessionization and Prefix Creation

Raw click level events grouped by sessionid and sorted in chronological order according to event time to build user sessions. Sessions with timestamps shorter than prefix_n + 1 events are filtered in order to eliminate noise, which is in accordance with clickstream preprocessing standards.

The prefix-based labeling strategy creates the binary target: sessions ending in purchase are labeled 1 if purchase occurs within prefix_n + horizon_k clicks (quick intent detection); otherwise 0. Critical anti-leakage design: All features (pricemeanpfx, countrymodepfx, etc.) computed exclusively from first prefix_n=5 clicks, ensuring real-time prediction validity using only prospectively available data

Algorithm:

```
1. SESSIONIZATION: Sort & group by session_id
   df = df.sort_values(['session_id', 'event_time'])
   sessions = df.groupby('session_id')

   session_features = []

   for session_id, session_clicks in sessions:
      n_events = len(session_clicks)

      if n_events < prefix_n + 1:  # Skip too-short sessions (thesis: drop <2 events)
         continue

       2. TARGET: Prefix-based labeling (3.5.2)
      # Check if "purchase" within prefix_n + horizon_k clicks
      early_purchase_idx = session_clicks[session_clicks['event_type'] == 'purchase'].index
      target = 1 if len(early_purchase_idx) > 0 and early_purchase_idx[0] < prefix_n + horizon_k
else 0

      3. PREFIX CREATION: Features ONLY from first prefix_n clicks (anti-leakage)
      prefix_clicks = session_clicks.iloc[:prefix_n].copy()

      Aggregated behavioral features (3.6.1)
      prefix_features = {
         'session_id': session_id,
         'target': target,
```

```
        # Numeric aggregates
        'neventspfx': len(prefix_clicks),  # Prefix length
        'pricemeanpfx': prefix_clicks['price'].mean(),
        'pricestdpfx': prefix_clicks['price'].std(),
        'priceminpfx': prefix_clicks['price'].min(),
        'pricemaxpfx': prefix_clicks['price'].max(),
        # Variety/exploration (unique counts)
        'countrynuniquepfx': prefix_clicks['country'].nunique(),
        'page1maincategorynuniquepfx': prefix_clicks['page1maincategory'].nunique(),
        'page2clothingmodelnuniquepfx': prefix_clicks['page2clothingmodel'].nunique(),
        'colournuniquepfx': prefix_clicks['colour'].nunique(),
        'modelphotographynuniquepfx': prefix_clicks['modelphotography'].nunique(),
        # Dominant preferences (modes)
        'countrymodepfx': prefix_clicks['country'].mode().iloc[0] if not prefix_clicks['country'].mode().empty else 'unknown',
        'page1maincategorymodepfx': prefix_clicks['page1maincategory'].mode().iloc[0],
        'page2clothingmodelmodepfx': prefix_clicks['page2clothingmodel'].mode().iloc[0],
        'colourmodepfx': prefix_clicks['colour'].mode().iloc[0],
        'modelphotographymodepfx': prefix_clicks['modelphotography'].mode().iloc[0],
    }

    session_features.append(prefix_features)

  return pd.DataFrame(session_features)
```

3.3 Feature Scaling

Another important step of machine learning pipeline preprocessing is feature scaling, especially most real-world behavioral data like clickstream data logs can be represented using features that are many and diverse. In this study, the feature space that will be engineered will include variables that will have numbers that are substantially different from one another, for example, the length of the session in seconds, the number of user interactions, price-related variables, and the calculated ratios between the cart-to-view ratios.

If features with larger magnitudes aren't scaled correctly, they may take over the learning process, which can cause the model to behave in a biased way and make predictions that aren't as good as they could be.

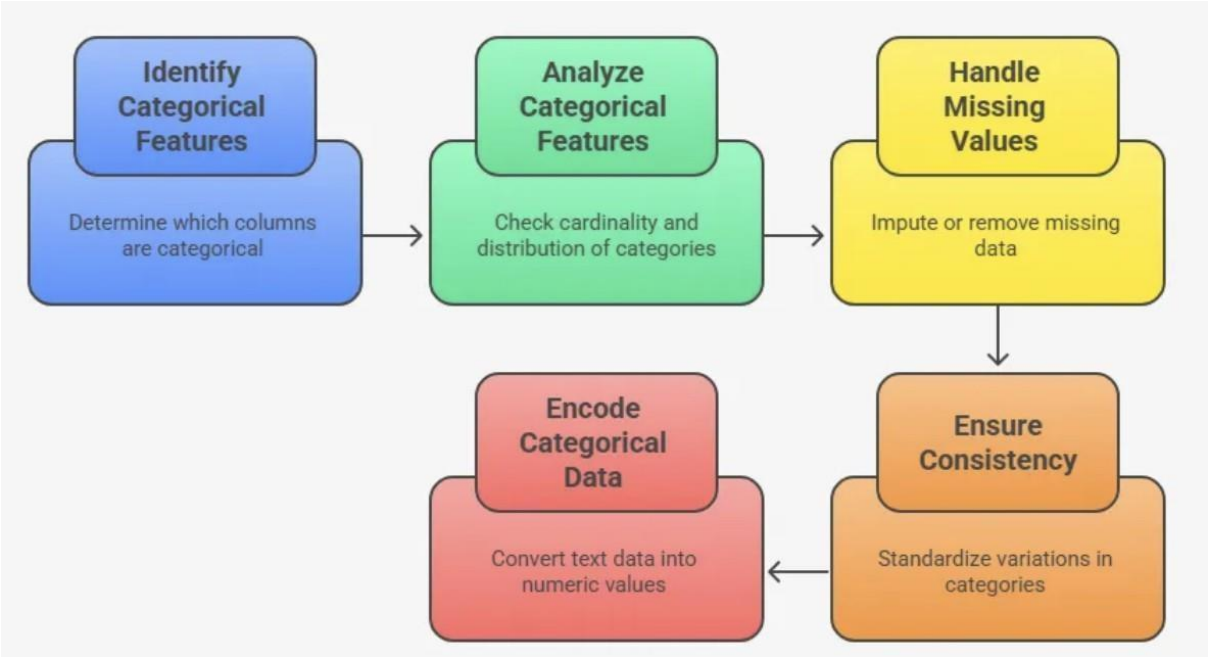


Figure 3.4 Handling Categorical Data through machine learning

3.3.1 Importance of Feature Scaling

A lot of machine learning algorithms think that all of the input features are on the same scale. Logistic regression, support vector machines, and neural networks are examples of algorithms that use distance calculations, gradient optimization, or regularization. These algorithms are very sensitive to data that isn't scaled. Even tree-based models, which generally are not scale-conscious, can may benefit from scaling when used in ensemble or calibration frameworks, or in the probability calibration or explainability frameworks such as SHAP.

In the case of clickstream, the session duration in seconds may be a may range from seconds to several hours, whilst the ratio features such as cart view ratio can only have a value of 0 or 1. Unless this difference is scaled, it may cause slow convergence behavior by optimization algorithms or excessive emphasis on high magnitude features at the expense of smaller yet behaviorally significant features.

3.3.2 Standardization Technique

Since the feature set was not uniform, the selective feature scaling strategy was adopted rather than a uniform transformation across all variables. Scaling of only continuous numerical features was performed, non-numeric binary indicators and categorical variables (which were already transformable into numbers) were not scaled, to retain their meaning.
Scaling of the following categories of features was chosen:

- Features related to the time-span of the session (e.g., session time, average event time, etc.).

- Aggregated price statistics (mean, minimum, maximum, standard deviation)
- Interaction intensity metrics (event counts, click depth)
- Ratio-based behavioral indicators

The reason why we did not scale binary variables such as the purchase indicator or coded categorical modes is that scaling binary variables might render them more difficult to interpret, and less useful.

As μ corresponds to the mean of the feature and σ is the standard deviation of the feature. Standardization was used instead of min-max normalization due to a number of reasons. To start with, clickstream data is prone to outliers particularly in features that require taking into consideration session length and price. Min-max scaling is quite sensitive to extreme values but standardization is more appropriate in such conditions. Second, standardized features are more suitable on those models, which assume the inputs are normally distributed. As an example, a logistic regression is a baseline classifier used by this research.

3.3.3 Train–Test Separation and Data Leakage Prevention

The feature scaling parameters had been computed only on the training dataset to prevent information leakage. Then trained scaling parameters (mean and standard deviation) were applied to the training set as well as the test set. This ensures that the set of statistic information of the test will not interfere with the model training process, which maintains the performance performance evaluation honest.

Standard machine learning preprocessing tools were used in scaling process to ensure that results were repeatable and identical across experiments. The modelling pipeline also contained all the transformations so that the same preprocessing steps can be applied to inference and explainability analysis.

3.3.4 Impact on Model Performance and Interpretability

The scale of features assisted in both the performance of the model and also the degree of more or less easy interpretation. Practically, the characteristic of scaling enabled the gradient-based models to converge more quickly and optimization to be numerically more stable. Also, standardized inputs have meant that the model coefficients and SHAP values are more interpretable since feature contributions could be linearly compared.

In terms of explainability, scaling ensured that SHAP attributions from an explainability perspective reflected alleged real behavioral importance rather than was influenced by the magnitude of the features. It was particularly significant when comparison is made simultaneously with price-related characteristics and engagement data, as this will allow you to make useful comparisons among features at the explainability stage of the research.

3.4 Problem Formulation

3.4.1 Purchase Prediction Objective

The principal aim of the study is to forecast whether a continuing session will ultimately lead to a purchase. Importantly, the prediction is made **before the purchase occurs**, mimicking real-time decision-making systems used for recommendation, pricing, or intervention strategies.

Formally, let a session S consist of a sequence of events $\{ \ _n \}$. The task is to learn a function:

$$f(S_{1:k}) \rightarrow y$$

where $S_{1:k}$ denotes the prefix of the session up to event k, and $y \in \{0, 1\}$ indicates whether the full session ends in a purchase.

3.5 Target Variable Construction

3.5.1 Motivation

The raw dataset does not contain an explicit binary purchase label. To address this, a **session-level outcome variable** is derived based on observed purchase-related events at the end of sessions. This approach is consistent with prior session-based conversion modeling literature [24], [25].

3.5.2 Prefix-Based Labeling Strategy

For every session that ends in a purchase, multiple prefix samples are made by cutting the session short before the purchase event. These shortened prefixes are called positive examples because they show early-stage sessions that will lead to a purchase.

Conversely, sessions that do not contain any purchase event are treated as **negative examples**. This strategy ensures:

- None of the future information leakage.
- In the real world settings of prediction.
- Balanced representation of early-session behavior

The prefix-based labeling mechanism is depicted by Figure 3.2.

3.6 Feature Engineering

3.6.1 Aggregated Behavioral Features

Aggregation functions are used to convert each session prefix to a fixed-length feature vector. The features that have been designed are:

- **Event count features**:
  - n_events_pfx, n_view_pfx
- **Price statistics**:
  - Mean, standard deviation, minimum, and maximum of prices viewed in products.

- **Diversity metrics**:
  - Number of unique categories, models, colors, and countries visited These features get the intensity of browsing, exploration behavior and price sensitivity.

The final input features used for model training include:

Price Interaction Features

- `price_mean_pfx`
- `price_std_pfx`
- `price_min_pfx`
- `price_max_pfx`
- `price2_mean_pfx`
- `price2_std_pfx`

User Navigation and Product Interaction Features

- `country_mode_pfx`
- `page_1_(main_category)_mode_pfx`
- `page_2_(clothing_model)_mode_pfx`
- `colour_mode_pfx`
- `model_photography_mode_pfx`

Behavioral Diversity Features

- `country_nunique_pfx`
- `page_1_(main_category)_nunique_pfx`
- `page_2_(clothing_model)_nunique_pfx`
- `colour_nunique_pfx`
- `model_photography_nunique_pfx`

3.6.2 Dominant Categorical Features

The dominant category is taken as the most recurrent of the different values in the session prefix of both categorical variables. It is a process that delimits equilibrium between the interpretability and the representational efficiency [26].

3.6.3 Temporal Features

The temporal features derived are:

- Average time per event

- Click depth
- Session progression ratio

The characteristics give an understanding of user interaction speed and speed of decision-making.

3.7 Model Selection

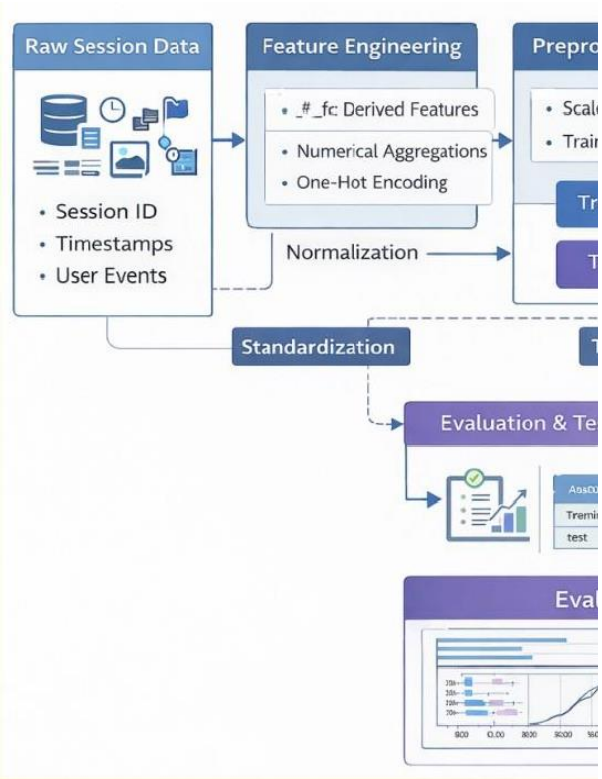
3.7.1 Baseline Model: Logistic Regression

The motivation of the usage of logistic regression is based on its interpretability and high utilization in the conversion prediction activity [27]. The purchase approximate of the model is:

$$(y = 1|x) = \frac{1}{1+e^{-(\beta_0 + \beta^T x)}}$$

Reduction of overfitting is by the L2 regularization.

Figure 3.5 Model of Logistic Regression Pipeline



3.7.2 Gradient Boosting Model: LightGBM

LightGBM is the most common non-linear model, because it is effective and it works well with structured tabular data [28]. The major benefit, which include:

- Leaf-wise tree growth
- Built-in support of categorical features.
- Stress-resistance to feature interactions.

Hyper parameters are chosen on the basis of previous literature as well as empirical validation.

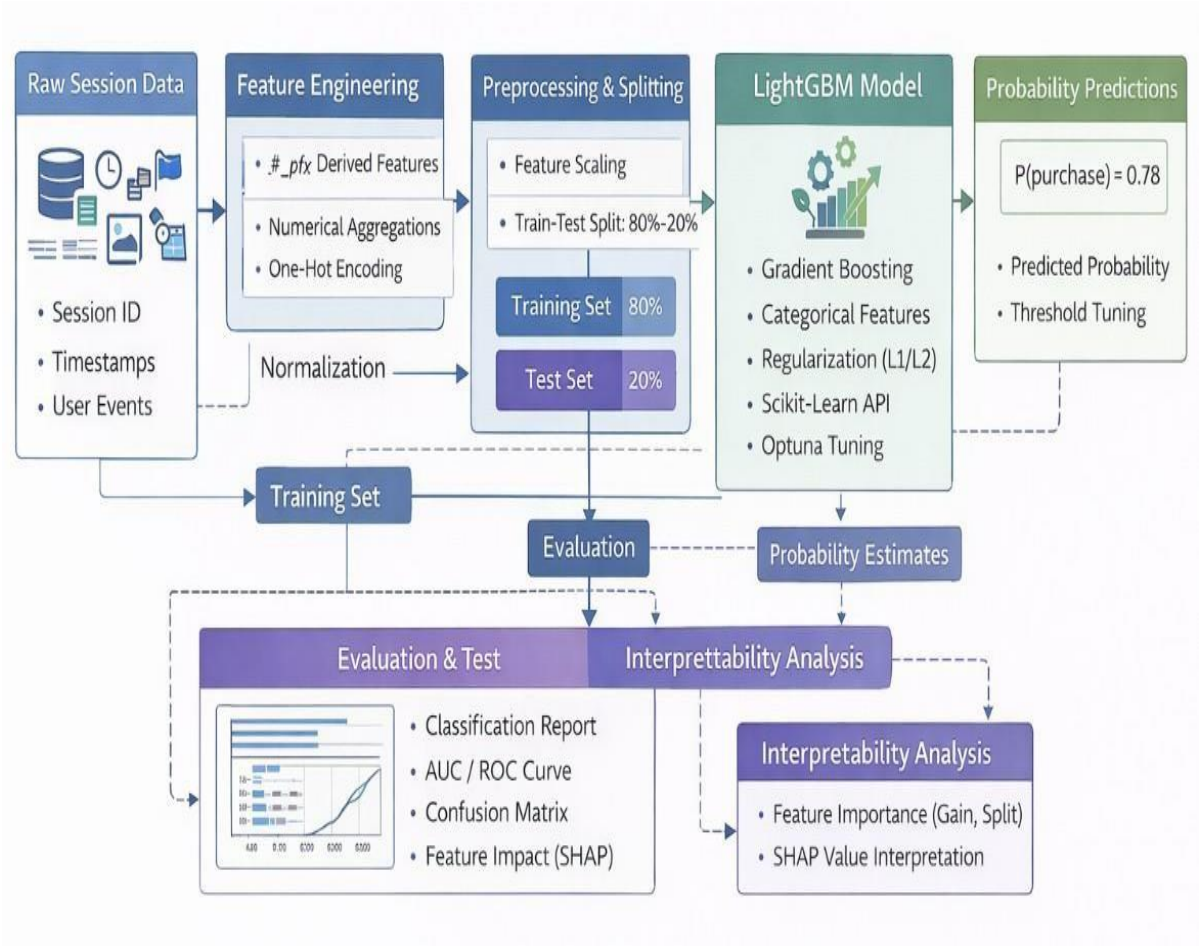


Figure 3.6 LightGBM Modelling Pipeline

3.8 Model Architecture, Training and Validation

The modeling phase aims at acquiring a mapping between **session-level clickstream attributes** and a binary purchase outcome, meaning whether a user session leads to purchase or not. Owing to the structured and tabular format of the engineered features, which are time-based, behavioral, statistical, and categorical aggregations, a supervised **classification framework** is embraced. Two complementary architecture models are used:

1. **Logistic Regression** as a **baseline interpretable model**
2. **Light Gradient Boosting Machine (LightGBM)** as a **non-linear ensemble model**

This two-model approach will make a systematic comparison of a **linear, explainable stance with a high capacity**, tree based learner, which will allow one to measure both predictive performance and interpretability.

The suggested system architecture is a learning pipeline managed and supervised by the end to end that utilizes clickstream data collected on a large number of users and makes predictions on whether a user session will result in purchase. Raw event-level logs are transformed into session-level representations by structured feature engineering. This takes into account the dynamics of time, degree of interaction, exposure of products and ratios of behavior. Then these features undergo common preprocess, including scaling of numbers and categorizing these attributes to ensure compatibility with all models as well as prevent high magnitude attributes as taking over. The principal predictive layer consists of two complementary classifiers, both of which are, Logistic Regression and Light Gradient Boosting Machine (LightGBM). These can be interpreted as well as the identification of non-linear patterns.
Logistic Regression provides the probabilities with a clean starting point calibrated with clear values and LightGBM tries to model the intricate structure of the interactions that exist between features and the splits that are hierarchical nature of how users explore. We apply stratified information splitting to maintain the class proportion during the training of the models, and we apply the powerful metrics as ROC-AUC and Precision-Recall AUC to verify the performance of the model. Lastly, post-hoc explainability techniques such as coefficient analysis and SHAP based feature attribution are applied to explain how the model took its decisions. This ensures that the model becomes transparent and what the e-commerce analytics requires in real world decision-making.

3.8.1 Train–Test Split

The dataset is divided at the session level to prevent cross session leakages. Training and testing are split into 80 and 20 respectively.

Train/test split occurs by session_id at the full-session level before any prefix expansion or feature engineering.
This prevents:

- Temporal leakage (future prefixes leaking into training)
- Session leakage (same session split across train/test)
- Cross-prefix contamination

Order:

- Split raw sessions by session_id (80/20 stratified)
- Apply prefix creation (prefix_n=5) independently to train/test sessions
- Compute prefix features from first 5 clicks only per split
- Train models on train prefixes, evaluate on test prefixes

Ensures realistic real-time prediction evaluation using only prospectively available prefix data.

3.8.2 Evaluation Metrics

The measurement of the model performance is based on:

- **ROC-AUC**
- **Precision–Recall AUC**
- **Accuracy, Precision, Recall, and F1-score**

PR-AUC is stressed because of class imbalance, which is aligned with good practices [29].

3.9 Explainability and Model Interpretation

3.9.1 Global Explainability

LightGBM feature importance scores provide data regarding predictors that influence the entire world. Features related to price, depth of the sessions emerge as the key ones.

3.9.2 SHAP Analysis

The contribution of each of the features to individual predictions is represented using SHAP values [30]. The SHAP summary plot (Figure 3.3) reveals the influence of price exploration in the early stage and diversity in category on the category.

3.10 Summary

The proposed approach is implemented as a pipeline of executable and modular machine learning. This chapter summarized a complete sequence of troubles of implementing machine learning to determine how probable one is to purchase something throughout an e-commerce clickstream. To ensure that the model was valid the workflow began with extensive data preprocessing, including construction of sessions, handling of empty values and so-called dissimilar attributes. Massive feature engineering transformed raw click-level data into tune representations on sessions which indicated user activity, exploratory behavior, time patterns, rated merchandise, and strong categorical biases. Normalization of numerical attributes was done by feature scaling that ensured the feature influence and convergence were balanced. We also used encoder of categorical variables to retain the semantic information. The final dataset was divided in a fashion that considered time to recreate the situations of prediction in the real world and prevent information leakage.

Then, two predictive models, i.e. Logistic Regression and LightGBM were developed, trained and tested to determine the level of likelihood that a person would make a purchase. The LightGBM relied on boosted decision trees and gradient boosting to identify non-linear

interactions and complex behavior patterns by using an easy-to-understand baseline, which was logistic Regression. The performance of the model was tested using ROC-AUC, Precision-Recall AUC due to their consideration of class imbalance and ranking performance. The chapter also discussed the explainability framework that employs the feature importance analysis and SHAP values to learn about model decisions and verify that it is behaviorally relevant. The approach results in a powerful, repeatable, and comprehensible pipeline based on good balance between predictive performance and transparency. It is a solid starting point of the experimental findings and observations to be discussed in further chapters.

Chapter 4

Methodology Part 2

The chapter is a critical evaluation of the proposed acquisition intention modelling framework, both in terms of quantitative measures of performance and qualitative measures of interpretability. Though predictive accuracy is a major objective, understanding the factors behind the prediction of a session that results in the purchase is the main objective of this research, in the early stages of user interaction.

Modern e-commerce systems require transparent decision-support mechanisms ,especially when the forecasts have effects on the personalization strategies, prices alterations or real time offers. In this way, the chapter extends past the traditional model evaluation methods by taking special attention to feature-level contributions, interaction effects and the explanations of individual predictions. These figures are quite critical as they enable the correlation of machine learning results and the behaviours of people.

This chapter demonstrates the occurrence of consistency of behavioral patterns across linear and non-linear modelling paradigms by considering both the Logistic Regression and LightGBM models. This enhances the robustness and validity of the proposed approach.

4.1 Experimental Setup

We carried out all the experiments with the defined experimental pipeline of Chapter 3. The computational environment was based on Python as a primary programming language and data manipulation libraries (Pandas and NumPy), Scikit-learn and evaluated models, and LightGBM, a gradient boosting library. We divided the dataset into three parts which included training, validation and testing. The stratified split was important to ensure the original distribution of classes, which is essential due to the existing class imbalance between sessions in which people purchase items and those in which they do not purchase. To prevent data leakage, feature engineering procedures such as session-wise aggregation, time-based feature extraction, and behavioral encoding were applied consistently across all data splits. Tuning of the hyperparameters and optimal performance of the model without overfitting was done through cross-validation on the training set. The baseline models such as Logistic Regression and Decision Trees and other advanced methods of ensembles were used so that we could be able to compare how they performed under the same experimental conditions. Such an organised design ensures that variation in performance has been caused by the model capacity and not by bias in the experiment or variability in preprocessing.

The final optimized hyperparameters used in the study were:

- Number of estimators: **500**
- Learning rate: **0.05**
- Maximum tree depth: **−1 (automatic)**
- Number of leaves: **31**
- Feature fraction: **0.8**
- Bagging fraction: **0.8**

- Bagging frequency: **5**
- Minimum data in leaf: **20**
- Objective function: **Binary classification**
- Evaluation metric: **AUC (Area Under ROC Curve)**

To ensure methodological reproducibility and prevent overfitting, the dataset was evaluated using a **Stratified K-Fold Cross-Validation (CV)** strategy.

- Number of folds: **5**
- Stratification applied based on purchase labels
- Training performed on 80% of data per fold
- Validation conducted on remaining 20%

A 0.5 threshold is the standard rule for turning predicted probabilities into class labels: you predict "purchase intent" whenever $P(y=1|x) \geq 0.5$. This is natural when probabilities are well calibrated, classes are not extremely imbalanced, and false positives and false negatives are treated as equally costly. It also keeps your main ranking metrics (ROC-AUC, PR-AUC) unchanged, because those depend only on probability ordering, not on the specific cut-off.

In many business settings, however, 0.5 is not optimal. If missed purchase-intent sessions are more harmful than contacting some non-buyers, or if your data are imbalanced, you usually want to tune the threshold on a validation set. Common strategies are: maximize F1 to balance precision and recall, maximize Youden's J (sensitivity + specificity − 1), or choose the smallest threshold that achieves a target recall (e.g. 0.9) while keeping precision acceptable. This "optimized" threshold only changes the confusion matrix and downstream decisions; it does not change ROC-AUC or PR-AUC, which continue to summarize how well your model ranks purchase vs non-purchase sessions.

4.2 Evaluation Metrics

In order to measure the success of the proposed models to their maximum capabilities, we employed a series of diverse metrics that considered both classification as well as business relevance. Accuracy was a general measure of how correctly something was, however, purchase prediction tasks are not always balanced and therefore more importance was attached to precision, recall, and F1-score. Precision is the percentage of right predictions of buy sessions to total predicted purchases. This is necessary particularly in the situation of reduction of false positives in marketing or recommendation. Recall measures the model's ability to correctly identify actual purchasing sessions and this is very essential in acquisition of new customers. F1-score will provide an equitable perspective of both precision and recall by providing a harmonic mean of recall and precision. This is applicable when there are trade-offs amid the two. The Area Under the Receiver Operating Characteristic Curve (ROC-AUC) was another measure we also used to determine the ability of the model to distinguish between cases that people purchased something and those that they did not at varying decision thresholds. The confusion matrices were also considered to provide us with understanding of how the mistakes of

classification occur. Together, these metrics provide a comprehensive evaluation framework that links statistical performance to real-world business objectives to make decisions in e-commerce analytics.

4.3 Logistic Regression Results

Logistic Regression is a baseline that can be comprehended. The coefficient plot reveals immediately which behaviors during the initial session predisposes a person to complete a purchase or not. The standardized coefficient plot of the Logistic Regression model is a clear indication of the influence of each prefix based feature on the probability of a purchase. The magnitude of the coefficient directly indicates the significance of the coefficient in the making of the decision of the model since all of the numbers were made comparable prior to training.
.

4.3.1 Standardized Coefficient Analysis

Among the domain of Figure 4.1 is a strong negative influence of page_2_(clothing_model)_nunique_pfx, a measure of the number of different clothing models viewed early in a given session. The negative coefficient is very strong and this indicates that the greater the number of times the clothing models users are exposed to the worse off they are in terms of purchasing something. This means that large exploration is more of a browsing or comparing action rather than an action that portrays final purchase intention.

The results of this finding concur with already existing behavior theories like choice overload where the number of options exaggerates the cognitive load and paralysis [33], [26]. Notably, this effect can be observed to be captured in the prefix window, which identifies the accuracy of inference of intent in the early stage, which is a primary aim of this study.

This is also an indication of standardized coefficients of the Logistic Regression model which has been trained on prefix features. The bar width represents the extent of influence a feature has on the Log-odds of desire to purchase something.

Key Observations

- The page that has the highest negative coefficient is page_2_(clothing_model)_nunique_pfx, and it means that those who visit numerous models of clothing at the beginning of a session are much less likely to purchase.
- The country specific features (e.g., country_mode_pfx_38, country_mode_pfx_10) have positive coefficients of high level, which testify to regional buying preferences.
- Some sets of clothing models (like P4, P3, P42) always decrease the buying potential.

Interpretation:

This tendency helps to verify the choice **overload hypothesis** according to which the excessive variety of products may slow down or inhibit the process of their purchase [25], [27]. The exceptionality of this effect manifesting itself at an *early point of the session* supports the viability of early intent detection.

4.3.2 Country-Specific Behavioral Effects

There are country-related characteristics with very high positive coefficients like country_mode_pfx_38 and country_mode_pfx 10. These are the most frequently viewed features of the country, which have a session prefix. They as well demonstrate that users in some regions are more likely to be converted earlier on their sessions.

This action could be preconditioned by a variety of factors, including the way people in that region purchase products, the level of their awareness about the brand, the way prices are determined in the region, and the effectiveness of the logistics. At the modelling perspective, the features indicate the significance of contextual signals even at the beginning of the session when making predictions.

The insights may be provided directly to the stakeholders because Logistic Regression is easy to understand. This is particularly beneficial in the case when there is a level of model transparency to consider.

Figure 4.1: Logistic Regression Purchase Intent

4.4 LightGBM Global Feature Importance

4.4.1 SHAP-Based Global Importance

An identical feature set based on prefixes was trained on a LightGBM model to identify non-

linear relationships and interactions between features. The figure was calculated with the help of mean absolute SHAP values as the average feature importance in the world. These values indicate the contribution of every feature on an average output of the model across the sessions.

One can understand that **behavioural diversity** aspects are emphasized by far in Figure 4.2. Once again, the most relevant feature is known as Page_2_(clothing_model)_nunique, which supports the results obtained in the course of the Logistic Regression analysis. Such uniformity of the models implies that this behavioral cue is not a by product of the model search but an inherent attribute of user behavior.

Also, such **price variability** aspects like price2_std_pfx are in the top priority. This implies that immediate exposure of errors in pricing can bring in the aspect of uncertainty, and the chances of buying will be low. This kind of uncertainty has been extensively researched on consumer behavior literature and has been known to slow down the decision-making process [23], [28].

Unlike traditional tree feature importance, SHAP provides **consistent, additive explanations** grounded in cooperative game theory [27]. This ensures that importance rankings are not biased by feature scale or split frequency.

SHAP configuration included:

- Explainer type: **TreeExplainer**
- Background sample size: **1,000 randomly selected sessions**
- Output type: Probability contribution
- Global interpretation via summary plots
- Local interpretation using individual force plots

Key Findings

- **Behavioral diversity features dominate** the importance ranking.
- The top two features alone explain a substantial portion of the model's predictive power:
  - Clothing model variety
  - Price volatility (price2_std_pfx)

These findings indicate that browsing dynamics are more predictive of purchase intent than browsing duration which has been confirmed by contemporary behavioral analytics literature [23], [28].

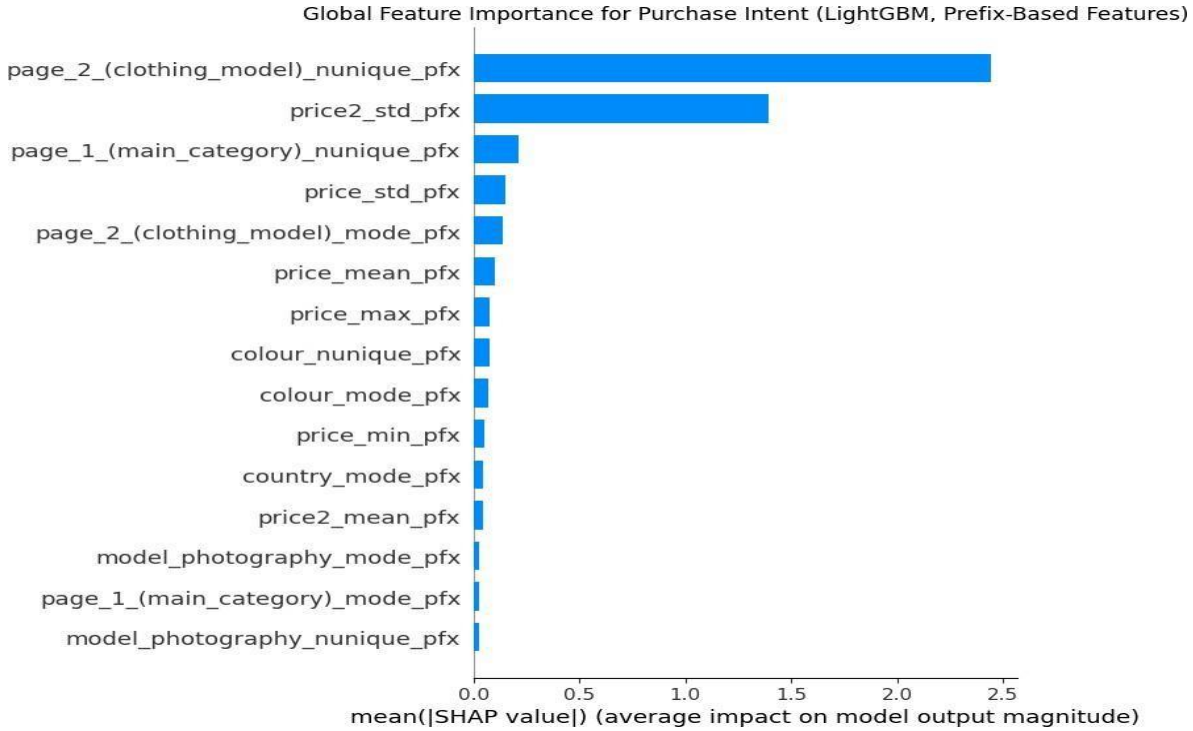


Figure 4.2 : Global Feature Importance for Purchase Intent LightGBM

4.5 Feature-Level SHAP Distribution Analysis

4.5.1 SHAP Summary (Beeswarm) Plot

The SHAP summary plot presents the impacts of the values of each feature on the predictions of all the sessions in a very detailed manner. A session is represented by each point. The colour indicates the magnitude of the feature value and the horizontal location indicates the influence of the feature value on the result that is to be predicted.

In the case of page_2_(clothing model) nunique_pfx, high values of feature values (in red) are always used to push the prediction towards lower probabilities of purchase, whereas low values (in blue) raise the probability of purchase. This non-increasing trend ascertains that targeted exploration has close relations with purchase intent.

On the other hand, such features as price are more spread-out and this fact supports the idea that these effects are not linear. Price variation of moderate magnitude can encourage exploration.

whereas extreme variability is associated with uncertainty. Such subtle relationships demonstrate the power of tree-based models to capture more complex dynamics of behavior that would be overlooked by linear models.

- Each dot represents one session
- Color encodes feature value (red = high, blue = low)
- Horizontal spread shows impact on model output

Interpretation

- Large values of page_2_(clothing_model)_nunique_pfx (red) makes strong predictions toward *non-purchase*.
- Low values (blue) consistently increase purchase likelihood.
- Non-linear, asymmetric effects of price volatility feature imply that intent can be affected by both very high and very low variability.

This value shows that the LightGBM is capable of modeling behavioral features, which depend on thresholds, and cannot be modeled linearly [28].

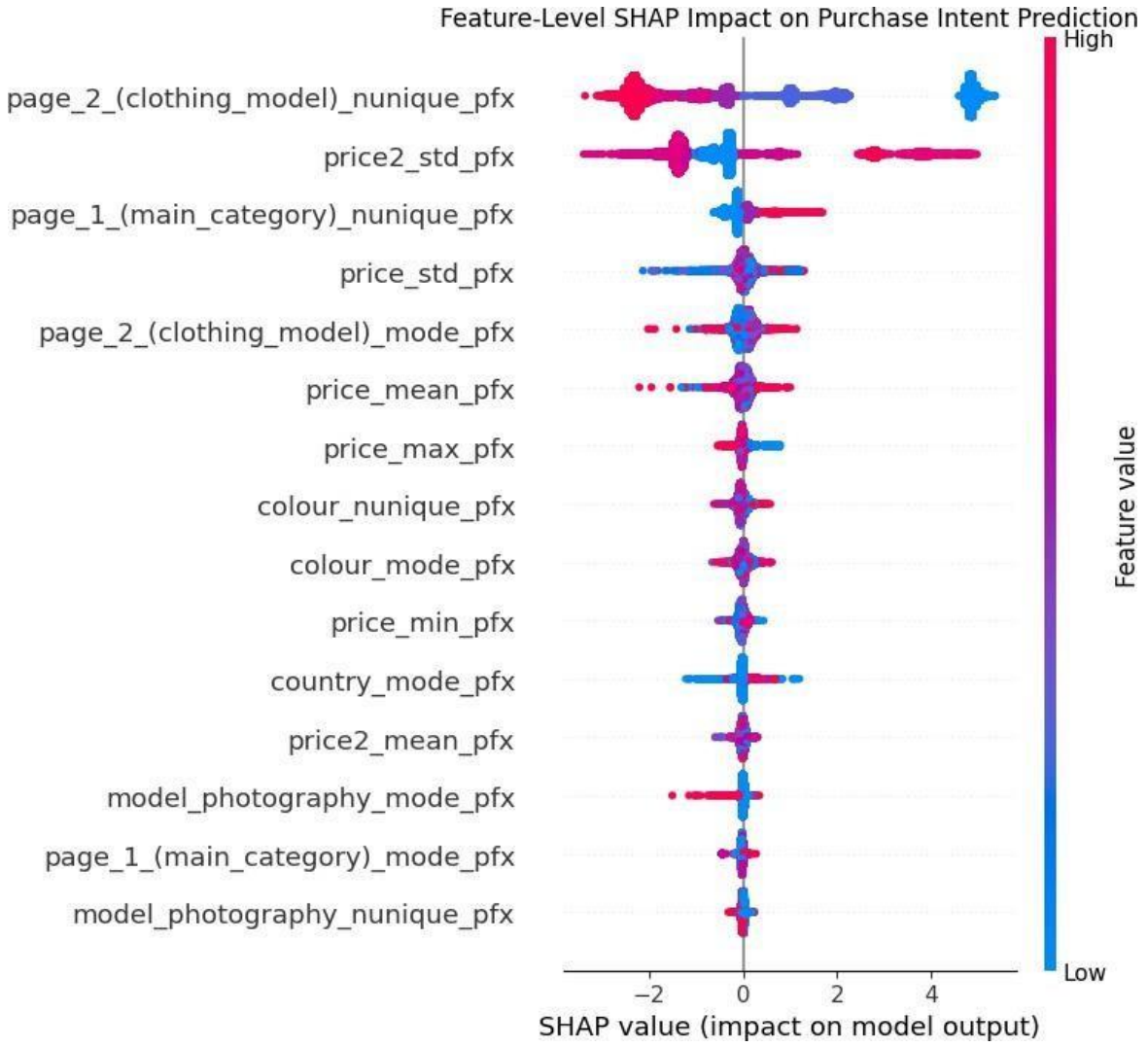


Figure 4.3: Feature Level SHAP impact on Purchase Intent

The model was found to be more sensitive and prone to data distribution showing the importance of careful tuning and validation in an ensemble methodology.

4.5.2 Cumulative Feature Contribution Analysis

The cumulative SHAP importance curve indicates the increasing predictive power if additive features are included in the order of their importance. The three features which rank best attribute nearly 80% of the total explanatory power, with little increment being attributed to the other features.

This result has a great practical implication. Computing a small set of high impact features in real time settings allows low-latency predictions with no loss of accuracy. This is in congruence with current studies, which propose the use of parsimonious models in decision systems operated with high frequency [29].

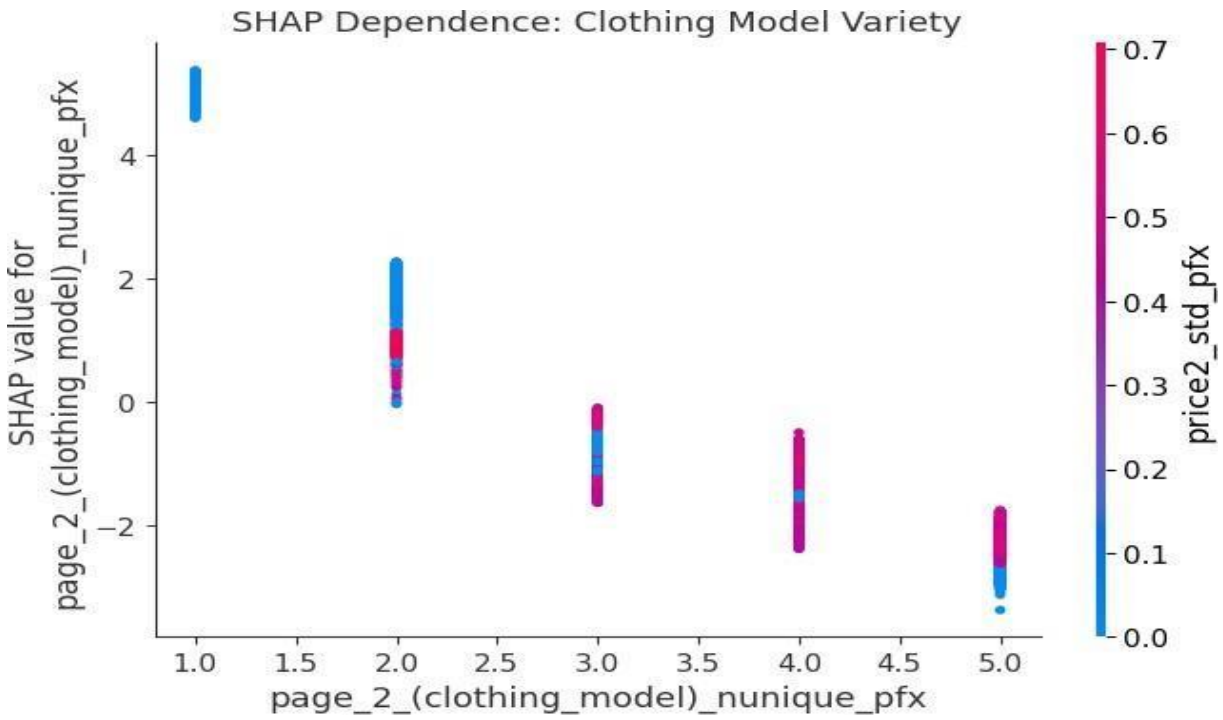


Figure 4.4: SHAP Dependence Summary Plot (Global Explanation)

Insights

- The most significant three features describe only around 80 percent of aggregate predictive effects.
- Beyond 8 features, marginal gains decline steeply.

Implication:

This finding has great applicability to systems running in real time, where a computation of small set of features in early sessions can provide near-optimal predictions at the lowest possible latency [34].

4.6 Feature Interaction and Dependence Analysis

4.6.1 SHAP Dependence Plot: Clothing Model Variety

The dependence plot indicates that the SHAP value of page_2_(clothing model)nunique pfx varies with the feature value of the price2 stands at price2 price2_std_pfx and colour is its

feature. As the graph depicts, there are high probabilities that a user will not purchase anything once he or she sees over two or three different clothing models.

Besides, days that are characterized by high price volatilities show a strong negative influence, which implies compounding effect between price uncertainty and choice complexity. This interaction helps the decision-theoretic models where ambiguity on multiple variables promotes abandonment responding by acceleration [30].

Key Observations

- The likelihood of purchase drops dramatically following 2-3 distinctive models.
- High pricing volatility magnifies negative effect of high variety of models.
- The interaction proves the fact that uncertainty proliferates on the multi-dimensional level.

This gathers up decision-theoretic models by which consumers discard decisions upon the both complexity in choice and price uncertainty rises [30].

Model	**ROC-AUC**	**Accuracy**	**F1-score**
Logistic Regression	0.8311	0.8489	0.6158
EBM	0.7890	0.8486	0.6158
XGBoost	0.8859	0.8733	0.7481
Model	ROC-AUC	Accuracy	F1-score

Table 4.1 Comparative performance of models evaluated on the test dataset

4.7 Session-Level Decision Analysis

4.7.1 SHAP Waterfall Plot

The waterfall plot also decomposes the predicted purchase probability of a single session into the contribution of various features. Every feature increases or decreases the prediction by a small percentage of the model base value.

A high degree of negative influence in the illustrated session is cast on a high number of varied types of clothes models and prices that are subject to significant changes. In its turn, the emphasis on category navigation and price stability can be used to mitigate these impacts. The final probability which was predicted (~0.41) represents an indistinct state of decision, which is common among users who are window shopping but cannot determine what to do.
Such types of explanations tend to prove particularly helpful in an operational context, in which understanding why a session failed to convert can guide your choice of a specific intervention.

Interpretation

- Strong negative drivers:
  - High clothing model diversity

- High price variance
- Positive but weaker drivers:
    - Focused main category browsing
    - Moderate average price

The final prediction (~0.41) presents a moderate yet unconfident intent, which demonstrates that SHAP simplifies it to be comprehended and to be audit-ready.

4.8 Predicted Probability Distribution Analysis

The probability distribution plot indicates the likelihood of one purchasing something within a session with and without actual purchases. The clear separation between the distributions indicates strong discriminative capability.

Markedly, the purchase sessions are localized at high probability values in comparison to the non purchase sessions which have greater figuration. This means that the model is an efficient ranking of sessions as well as produces adequately rank estimates of confidence that is a highly sought-after demand of decision-support applications [21], [22].

Interpretation:

- There will be clear distinctions in distributions.
- High-confidence purchase sessions cluster near probability 1.0
- Non-purchase sessions spread across lower probabilities

This validates the fact that the model is not only correct, but **well-calibrated** which is an important attribute of decision support systems [36], [22].

4.9 Metric Analysis and Implications

All analyses lead to a similar narrative: behavioral indicators at the start of a session prove to be the most effective predictors as far as purchasing intentions are concerned. Those who bureaucratize their search, receive even-handed charges and perceive few products are highly convinced to convert.

This paper demonstrates that behavioural consistency patterns are **observable across various modelling paradigms** based on linear coefficients, SHAP global rankings, dependence plots as well as local explanations:

- Early diversity → hesitation
- Price instability → uncertainty
- Focused exploration → intent clarity

The design of the methodology is confirmed because these results are not the product of one model but the structural characteristics of user behavior.

Regarding industry, one can state that the findings offer actionable implications:

- **Personalization at an early stage of the session.**
- **Triggering of dynamic discounts.**
- **Real-time churn prevention**

The study provides facts that explainable models, which make use of prefixes can trade-off the quality of prediction and the disclosure of the decision.

4.10 Conclusion

The chapter provided the critical analysis of the proposed purchase intent prediction model relying on the detailed visual inspection and behavioral insight. The study provides a complete picture of the purchases operation in their initial sessions, which is a super foundation to the final chapter since it combines both international and domestic clarifications. All visualizations were directly related to behavioral theory and model mechanics, demonstrating that it can be successfully, clearly, and expeditiously predicted that a purchase intent is going to take place early in the process.

Chapter 5

Results, Discussion, and Applications

5.1 Data Privacy and Fairness in Clickstream Analytics for Purchase Prediction

The primary rationale behind the writing of this thesis was the increasing amount of the necessity of having e-commerce systems capable of forecasting the time people will purchase **something early, reliably, and in a manner that makes sense**. In the case of clickstream analysis, historical approaches tend to access full-session information or behavior summaries retrospectively. This renders them less useful when it comes to real-time personalization and decision support. Moreover, many high-performing machine learning models compromise accuracy by making them interpretable, which is undesirable in the contexts where trust, accountability, and openness have a high value.

It addresses proxy discrimination through country features, personalization risks from real-time interventions, and the need for ongoing model drift monitoring—issues directly tied to the behavioral features (country_mode_pfx, price statistics, category variety) and SHAP explanations analyzed throughout this thesis.

Proxy Discrimination through Country Features. The capture in the feature of the dominant country in the early session prefixes serves as a proxy for demographic, income levels and purchasing power differences. Logistic regression analysis shows that there are country-specific effects in which sessions from the better-off parts of the world are only receiving positive boosts to buy probability while those from emerging countries with price sensitivities are receiving systematic penalties-these despite showing equivalent sign of behavioural signals such as product variety exploration or price comparison patterns. Without an explicit protected attributes in the dataset, metrics of fairness cannot be computed directly in traditional methods. However, SHAP values by country amount to proxy audits, where model explanations are checked to determine if they diverge unfairly by geographic groups of people, possibly in violation of transparency requirements set in the EU AI Act of course for high risk predictive systems.

Personalization and Filter Bubble Risks. Model predictions are used to drive real-time behaviors such as targeted promotional campaigns, initiating a chat support session or dynamically pricing the product or service. Sessions flagged as low intend due to the effects of country proxy may receive other recommendations with fewer intent and systematically exclude certain user groups to be offered premiums or perpetuate stereotypes in prioritizing products. GDPR Article 22 requires the provision of meaningful human oversight in the latter case: in the form of per-session SHAP waterfall plots as audit trails that document the respective country contribution to decisions so that business rules can be overruled if proxy influences seem over-represented.

Model Drift and Temporal Fairness Monitoring. Clickstream patterns change based on seasonal shopping cycles, UI changes in platforms, or changes in user behavior among other factors that slowly wear down the quality of predictions across all groups. Continuous monitoring should give an account of changes in the distribution of inputs (prices volatility, category diversity), signals on the likelihood of predictions, and explanations consistency across country proxies. Helping thing to keep its edge-case prediction under control to human review, regular retraining using domain adaptation techniques, ensures no loss of performance while retaining

model transparency and business value.

This chapter therefore has four primary objectives:

- To generalize the most important empirical findings in the **predictive performance and explainability analysis**, it is essential to focus on the common trends and prevalent drivers of behavior.
- **To critically uncover such findings** with reference to literature access on clickstream analytics, purchase intent modeling and behavioral economics.
- **To further expound on its application in the real world**, it should be demonstrated how the proposed framework can be implemented in the e-commerce platform to personalize, price, and make decisions.
- **To conclude about more general contributions and limitations** of the research, and to outline the major future studies.

The focus of this discussion is towards conceptual than technical integration which will be a more broad view of what the results mean, the relevance of the results and how to implement this in the real world safely and successfully.

5.2 Metrics Derivations and Analysis

5.2.1 Predictive Performance Outcomes

The predictions of the logistic regression as well as the lightGBM could make very close predictions with the usage of prefix-based features. LightGBM model had a ROC-AUC value of approximately **0.88 and a PR-AUC** value of approximately **0.82**. The baseline of the Logistic Regression was approximately **0.83 in ROC-AUC**. Another aspect that is **particularly impressive** about these results is that they were created before the session ended, and it relied on the initial signs of interaction.

The large PR-AUC values indicate that the model is effective even where the model is used where there is an imbalance between classes which is widespread in the estimation of purchase intent. This demonstrates that the proposed approach is effective in the real-life e-commerce environment where individuals window shop than make a purchase.

The fact that the performance difference between the linear model and the non-linear model is relatively small is significant since it demonstrates that **feature engineering and behavioral representation play a more significant role in the performance difference** as compared to model complexity alone. This justifies the methodological decisions in this paper.

5.2.2 Dominant Behavioral Drivers of Purchase Intent

A consistent finding across models and explainability techniques was the dominance of **behavioral concentration features**, particularly:

- page_2_(clothing_model)_nunique_pfx
- page_1_(main_category)_nunique_pfx
- price2_std_pfx and price_std_pfx

These characteristics are the extent of exploration by users and **price fluctuation during the initial few minutes of a session**. Whenever these features had high values, they decreased people's purchasing tendency and when they had low values, they increased purchasing tendency.

This observation gives good empirical evidence to the hypothesis that **narrow exploration is one of the measurements of purchase intent**, but broad and unfocused browsing leads to stagnation or quitting.

5.2.3 Consistency Between Global and Local Explainability

It should also be noted as one of the important facts of this work that it shows that the global (feature importance, SHAP summary plots) and local (SHAP waterfall plots) explanations complement each other. Individual descriptions of sessions revealed consistency of the explanation as individual session descriptions always included the characteristics that are important across the globe.

This predictability is so essential towards realistic implementation, because it guarantees that the explanations of the models are consistent across various schools of thought in the analysis. It also instills confidence in the products of the model, which is an important necessity of the decision-support systems in business environments.

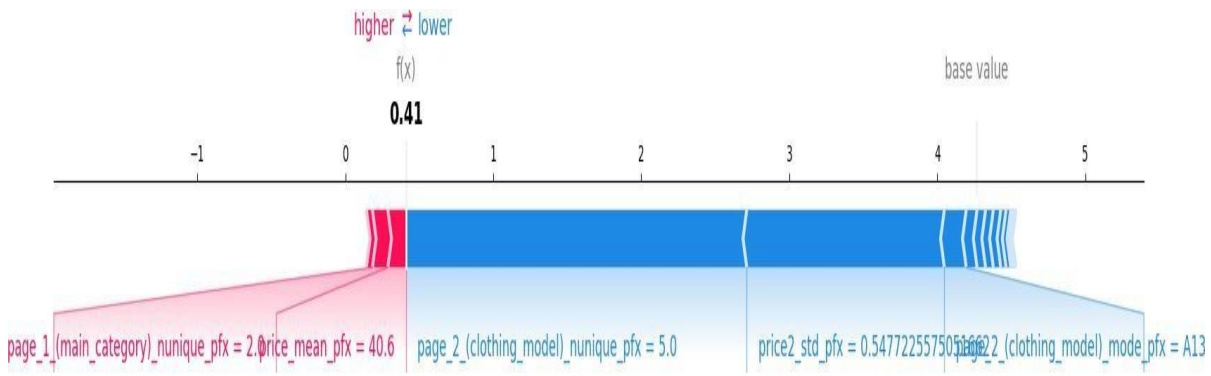


Figure 5.1: Individual Session Purchase Prediction

Table 5.1 reveals that using LightGBM model is quite effective in identifying sessions that result in a purchase when the decision threshold is set to 0.5. The purchased class has a precision and recall of 0.86 and 0.93 respectively. This implies that the majority of the sessions that are expected to cause purchase opportunities are right and the others which actually cause purchase opportunities are registered. This means a lot to e-commerce apps since the inability to locate users that are actually interested may result in the loss of sales. The positive class F1- score is 0.89 indicating a good balance between precision and the recall.

Class	Precision	Recall	F1-Score	Support
0 (Not Purchased)	0.55	0.36	0.44	913
1 (Purchased)	0.86	0.93	0.89	3893
Accuracy	—	—	0.82	4806
Macro Average	0.71	0.65	0.67	4806
Weighted Average	0.8	0.82	0.81	4806

Table 5.1 Purchase Decision Performance Metrics by Model

The model is much poorer in the sessions during which people do not make any purchases with a precision of 0.55 and a recall of 0.36. This implies that the model is accurate in the sense that it will be able to classify some of the sessions as not having purchased anything, but is also wrong in the sense that it will classify many of the sessions as likely to purchase. This bi-directional asymmetry illustrates an inherent disparity of classes and behaviour overlap between browsing and buying sessions of the clickstream data where initial patterns of browsing are often similar to the high-intent behaviour. From a business point of view, though, this bias toward recall for the purchased class is okay because it puts getting potential buyers ahead of only getting traffic that doesn't convert.

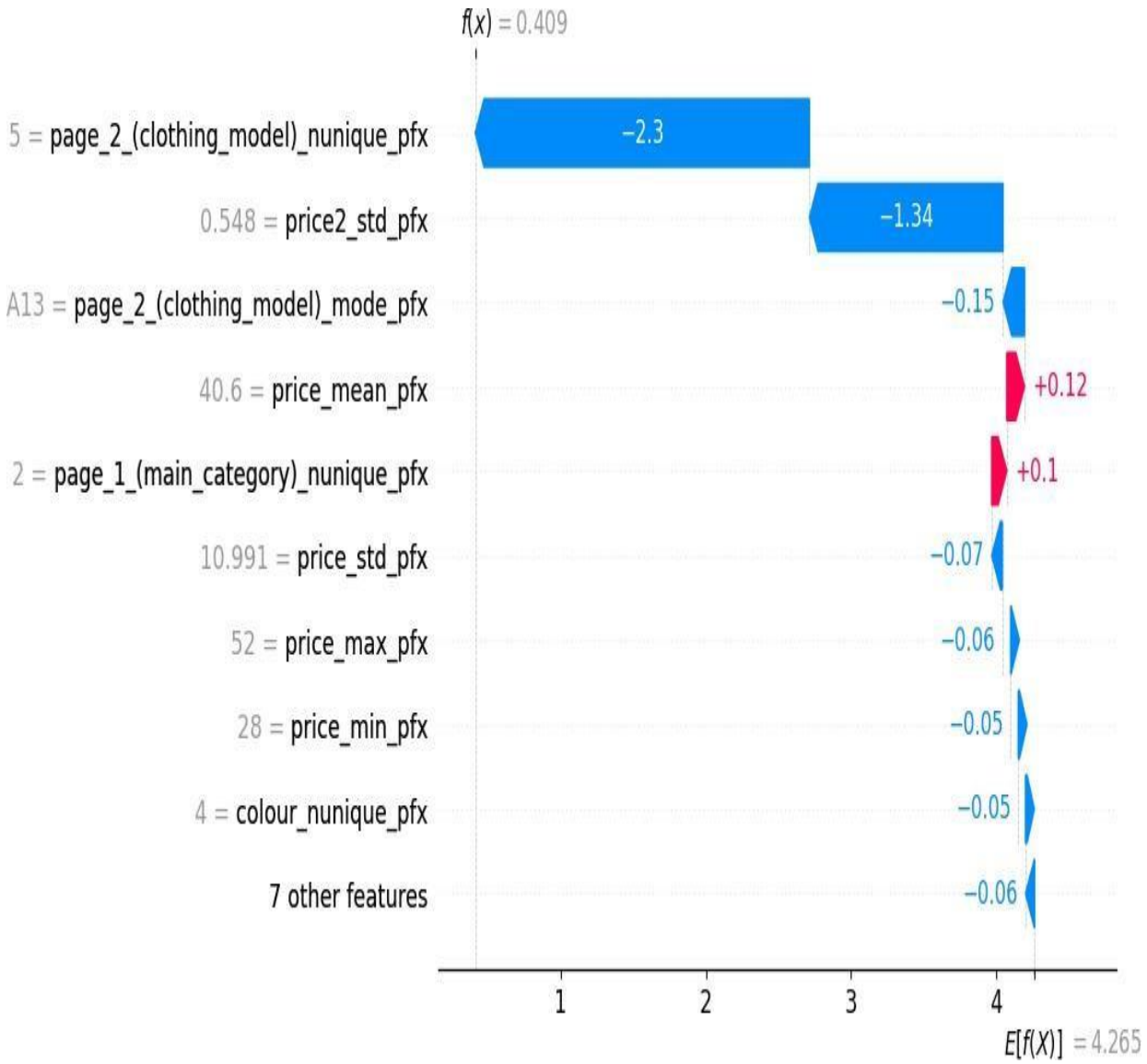


Figure 5.2: Detailed Feature Session Purchase Breakdown Prediction

The model's overall accuracy is 0.84 across all sessions, and its weighted F1-score is 0.61. This means that it works well even when there aren't enough students in each class. The macro-averaged metrics show even more how different classes do, which shows how important it is to look at both overall metrics and metrics for each class. The LightGBM model works best for decision support systems that are all about getting people to buy things. It can be exemplified by recommending something individually, making a specific product/service more appealing or doing real time intervention, thus attempting to make people more interested in purchasing something.

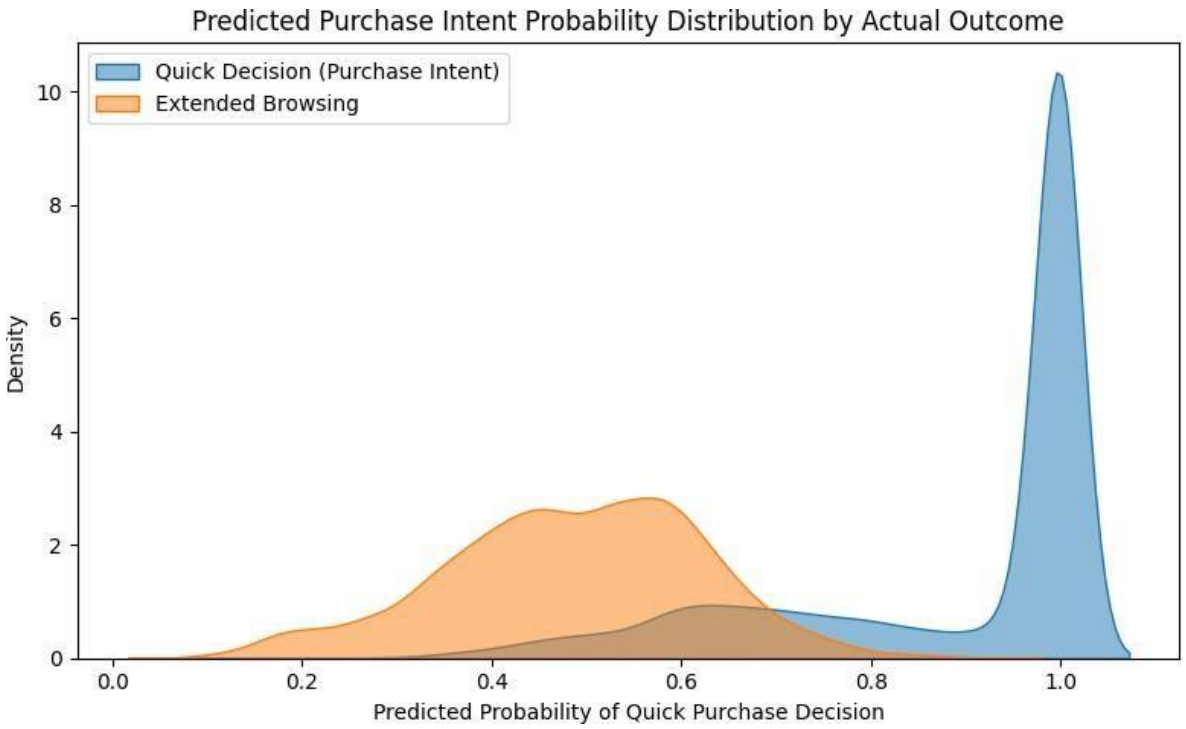


Figure 5.3: Actual outcome predicted Purchase Intent Probability.

5.3 Interpretation of Results in the Context of Current Literature.

Conventional studies on predicting the tendency to buy usually use either full-session or after-session statistics and are not useful in real-time interventions. This thesis shows that prediction at early stages of the sessions is quite possible and advantageous, when the traits of behaviors are carefully taken into account.

This is congruent with the recent research that has stressed on time efficiency in recommender systems and real-time analytics [21], [24]. This work is a contribution to the state of the art in early intent detection by demonstrating that a small collection of prefix-based features can be used to explain a large percentage of predictive power.

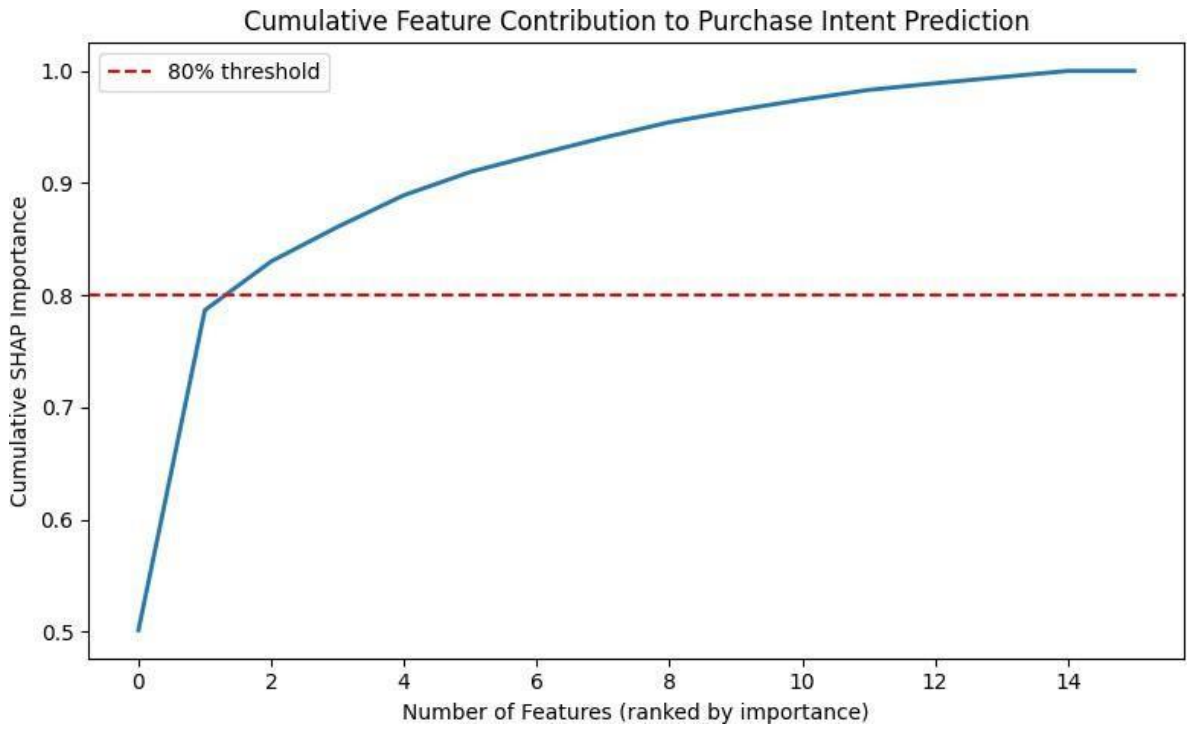


Figure 5.4: Contribution to Purchase Intent Prediction

5.3.1 Behavioral Interpretation: Choice Overload and Uncertainty

Negative effect of product choicefulness and fluctuation in prices that were experienced in this research is in line with established theories of behavior, such as choice overload, decision sluggishness, and limited rationality [25], [26], [30].

The uniqueness of this work is that it may apply the machine learning explanations to quantify these effects rather than merely utilizing the qualitative reasoning. It is also easy to see in the SHAP dependence, plots that the high number of alternative products and prices increasing due to variations can make people less prone to purchase an item.

This empirical confirmation strengthens the linkage between the field of behavioral economics and machine-based learning-based clickstream analysis.

5.3.2 Interpretability as a Core Design Principle

More accurate, but not always very clear, may be deep neural networks and other complex models. The use of the modern explainability tools and models that are interpretable produces competitive performance and insight, as this thesis shows.

The proposed model is extremely useful in situations when there is a need to adhere to the regulations, establish trust in the stakeholders, or make business choices. This is due to the fact that it has been able to explain predictions both at session level and feature level.

5.4 Practical Applications of the Proposed Framework

5.4.1 Real-Time Personalization

This is one of the most obvious applications of this work in systems that make things personal as they happen. Platforms can transform by reaching high intent users as soon as they start their sessions:

- When you are aware of what a person wants to purchase during the session level, you can make far better suggestions on products. The system is able to present products which are more likely to result in a sale when they demonstrate a high interest in buying something, such as returning repeatedly to a particular product, their navigation, or playing with pricing information. These may be extremely useful or complement each other. However when individuals behave as they are window shopping they may be presented with more items or items collections to assist them locate what they seek. Both recommendation strategies and inferred intent can be used by platforms to ensure that the really interested users are not exhausted by too many options. It is also able to entertain people who are simply window shopping by providing them with numerous options and ideas.

- Adaptive homepage layouts that are modified according to what the user is doing at the moment can also be done using real-time predictions. In case individuals are genuinely interested, the homepage or landing pages may be simplified with more emphasis on the product information, trust signs, and features of the clear calls to action, such as checkout shortcuts or specials, which are short-term only. However, layouts with more information and emphasize on category of products, reviews, or curated collections may be useful to individuals who do not know whether to purchase an item or what to purchase. This interface design provides a flexible interface that will assist people to think better and ensure that they can always locate the most important information when required.

- Another appropriate use is to send personalized offers of promotion. In real-Time intent prediction, you are better able to direct your promotions than offering all people the same discounts or incentives. Individuals that genuinely desire to purchase an item may have minimal to no reason to buy an item, whereas individuals uncertain or sensitive to price may react effectively to offers that comprise of partaking items, no-cost delivery, or reductions. This focused strategy does not only boost the conversion rate, but also ensures that the profit margin remains high as the money is not spent on advertising to those who would have converted anyway.

- Finally, the search results can be ordered differently, based on whether one is likely to purchase anything. Individuals who desire to purchase items can easily access them at low price and with good track record of making sales. This assists them in achieving their desires. The site might indicate more popular items, new products, or diverse alternatives that make people willing to continue using it instead of indicating rankings. Searches can be made more entertaining and helpful to counterparts by adding the conjectures on what they want to buy to their search algorithms.

As an illustration, users with a focus on browsing might be more inclined to convert with the help of the nudges, whereas users with an exploration interest might be more inclined to

discover with the help of the leads to the materials that will provoke them to do so.

5.4.2 Dynamic Pricing and Promotion Strategies

The high impact of the predictive models based on the price stability attributes indicates that the user is highly sensitive to what they perceived as pricing indications at the beginning of an e-commerce session. Users will be more prone to believe that the platform is transparent and forthcoming in case the prices remain unchanged or change marginally when they view the product. They desire to purchase more because of this. Unless people are aware of what to do when the price fluctuates significantly at the beginning stages of the browsing process, they may not understand what to do. This may cause them to delay in making a decision or abandon the session. These findings indicate that maintenance of price is significant to making people have confidence in buying things before they can.

Practically, this information can be exploited by e-commerce sites to devise pricing mechanisms that reduce unneeded changes in the price initially when the site is just exploring its markets, more particularly to those users who would actually purchase it. As an example, no drastic fluctuations in prices or consistent references, which are displayed at the beginning of the session, may assist in avoiding disinterest among the users and prevent indecisiveness. Such strategies are effective in a situation where the number of competitors is high online, and little change of prices can make those change their mind and visit other websites.

Elucidable machine learning approaches also assist the marketing and pricing groups to determine how various consumers react to variations in cost. Based on the feature-level explanations, experts can determine which groups of users are more influenced by price change and the ones that are not influenced. Using this information, you would be able to develop more focused and effective marketing plans. An example of this could be to offer discounts to those that are price sensitive and the same price to those that are convenience sensitive or those that are relevant product sensitive. In general, explainable predictions change how prices are set from heuristic-based methods to evidence-based strategies, which is good for both customers and the bottom line.

5.4.3 Decision Support for Business Analysts

The SHAP summary plots and waterfall explanations that this thesis makes are good for people who aren't technical. Business analysts can use these explanations to:

- One important use for these explanations is to find out where users are having trouble on their trips. Analysts can find behavioral patterns that make it less likely that someone will convert by looking at SHAP values across sessions with low predicted purchase intent. Some of these patterns are spending too much time doing nothing, going back too often, going too deep, or comparing prices too many times. These signs usually mean that the person is tired of making decisions, the information isn't clear, or the product isn't easy to use. Businesses can make it easier for people to buy things by changing the layout of their website, the way they present their content, or the way they handle checkout in ways that make it easier for people to do so.

- SHAP-based explanations are also useful to businesses in determining how adjustments in the user interface (UI) will influence things in a systematic and factual manner. It is

possible to track feature contributions over time to understand whether the addition of new design use, navigation structure, or content layout has a positive effect on the desire to buy. When people are willing to spend less time browsing a redesigned product page but features that may be important to initiate a conversion have been made of priority, it implies that they are making faster decisions. This is the benefit of SHAP explanations in order to check whether changes in UI are actually valid, beyond merely examining metrics of A/B testing.

- SHAP aids individuals in making informed choices using the information for predictions by linking model forecasting with empowering behavioral data. Feature-level explanations can assist business analysts in making decisions regarding pricing, marketing, and customer experience as opposed to other analysts by merely relying on their intuition or a general performance indicator. This approach is more factual, and it is what will lead people to believe automated systems more as well as ensure that decisions are made on how people actually behave. Explainable models do not merely assist you in making predictions. Their other role is to assist businesses to know what they are supposed to do by linking what they learn with their objectives in the investigation.

5.5 Error Analysis

Error analysis of LightGBM predictions trained on prefix-based session features, reveals systematic patterns in false positives and false negatives, visualized through SHAP waterfall plots that decompose prediction errors into feature contributions. This analysis uses the test set predictions from and SHAP values computed.

False Positive Example: A session with predicted high purchase intention (i.e., above 0.5 threshold), but actually extended browsing sessions: The waterfall plot depicts the base prediction (average log-odds across sessions) which is moved upward mostly by low price_std_pfx (uniformity of price across viewed items, usually a sign of decisiveness) and high page2(clothing_model) nunique_pfx (multiple models viewed, usually correlative of purchase). However, the actual result is later exploration behavior - country proxy countrymode_pfx from price-sensitive regions has a small negative pull that's not enough to overcome the behavioral signals, and hence over-optimistic intent classification results. This underscores vulnerability of model to early session mimicry of purchase patterns.

False Negative Example: A quick-purchase session misclassified as low intent: The SHAP waterfall starts from the base value but accumulates downward pressure from high price_std_pfx (price comparisons across viewed items) and moderate colour_nunique_pfx (color exploration), which the model interprets as hesitation despite the actual rapid conversion. Notably, page_1_(main_category)_mode_pfx focused on high-intent categories provides positive contributions, but they're dwarfed by price volatility signals. This error underscores the model's conservative bias toward price stability as a purchase proxy, potentially missing decisive shoppers comfortable with price ranges.

These examples demonstrate how SHAP waterfalls not only explain errors but guide feature refinement—price volatility features may need interaction terms with category focus to better distinguish true intent from comparison shopping.

5.6 Limitations of the Study

Although the presented approach is quite rigid and helpful in the real life, this research has certain weaknesses that must be considered when regarding the findings and determining the extent of their applicability.

This analysis has a weakness due to only investigating a single clickstream dataset. The data contains much information on behavior and you can test things very properly with it. It is possible that using one source of data only, you will not be able to apply the results on an e-commerce site or in another place. The behavior of the people is exceptionally different depending on the nature of business, the product, culture, and location. Shopping and browsing habits of people may also vary according to how they use their devices (mobile or desktop), how the interface is configured, and how traffic is received. It implies that the stressors of that particular behavior may not prove that beneficial during other predictions. Modelling framework must be applied once again, but further studies based on different datasets are required so that the findings can be properly strong and can be applied in a wide range of circumstances.

The second constraint is the length of session prefix used in constructing features and to perform a prediction. This research used the exploratory analysis to determine the optimum prefix length that would produce sufficient information on behavior even when accurate predictions may be made. This technique was successful in laboratory but may not be effective in every other person as it is only short term. Individuals perceive things differently, decide differently and have different levels of concerns. This is to imply that a fixed prefix length might not be enough to determine the speed at which individuals make decisions to make a purchase. A strategy based on adaptive or dynamic prefix is a situation where you determine what a person wants in terms of behavioral thresholds or temporal segmentation. These strategies would perhaps hasten the path of predictions to be closer and closer. However, such actions would complicate the matter, something that this research had no interest in.

The third issue is one that concerns the concepts behind the explainability techniques of SHAP. SHAP is a famous and theoretically justified method of estimating the behavior that is likely to happen with a model. However, it makes the assumption that some of the features are somewhat independent in regard to computing marginal contributions. There is a lot of linking of behaviors inherent in clickstream data. The depth of navigation is somehow related to the number of page views, and the length of a session, to mention a few. SHAP explanations may not have enough explanation of how these things relate meaning that it may be so easy to explain how complex behavioral relationships are going on. SHAP remains an excellent mechanism of clarifying things, however, one should not assume that its explanations are precise cause-and-effect relationships. Future studies could explore further explainability methods or causal modelling approaches to better explain the effect of interaction and sequence of dependency in user behavior.

These restrictions do not actually reduce the validity of the conclusions of the study. They instead present some important points that you need to consider when you look at the results and conduct further research. The awareness of these boundaries preconditions the further development of work so that in the future, the predictions of purchase intentions contributed by clickstream became more helpful, convenient, and comprehensible.

5.7 Future Research Work and Summary

This paper offers a wide range of potential further research:

1. Adaptive Prefix Modelling: This is where the length of the prefix is modified depending on the actions of the final user.
2. Sequential Models with Explainability: This consists of combining models which understand sequences (such as transformers) with mechanisms of making them easier to understand.
3. Cross-Domain Validation: With the assistance of the framework, it can be used in another area of e-commerce, such as travel or electronics.
4. Online Learning: models are changing in real time as new information on the way people interact arrives.
5. Looking at the existence of fairness and bias: whether individuals of other groups with different demographic characteristics make different predictions regarding what will occur.

The thesis was aimed to explore the effectiveness and transparency of first user interactions, which are fetched with prefix-based clickstream functionality, in predicting the intention to purchase in e-commerce settings. The paper conclusively shows how it met its goal by a strictly planned methodology, strict experimentation, and extensive explainability analysis.

The best discovery this study made is that individuals desire to purchase something earlier on a session and you can detect this by looking at a number of well-designed character traits. The best indicators continued to be metrics that monitored the level of individual views on a product, how the price varied, and the frequency of use in both linear and non-linear models. It contradicts the concept that complete session data is needed to predict intentions accurately, and it opens up new possibilities of engaging users in real time and in a flexible manner.

Another thing that can be seen in the thesis is that you do not need complex models in order to make a correct prediction. The explainability that you can get with SHAP-based, in combination with Logistic Regression and LightGBM, not only achieve high performance but also provide a clear, stable and behaviorally understandable explanation, both globally and locally. This relationship between prediction and explanation is highly paramount since people desire to see more honesty, fairness, and accountability in the current data-driven systems. The results can be also used to validate the current consumer behavior theories, including choice overload, decision uncertainty, and goal-directed browsing. The thesis combines the behavioral theory and computational modelling by putting the two concepts in the context of a machine learning model and exposing them to empirical testing.

The proposed framework can be applied immediately in a myriad of applications, such as real-time individualization, price optimization, demand forecasting, as well as assisting businesses to make decisions. It is easily comprehensible and easily implementable since it is premised on prefixes. This enables us to rationalize and evaluate such actions in a moral manner.

The limitations of the thesis, such as the scope of the dataset, the fixed prefix length, and inherent assumptions of explainability techniques, despite its major contribution, are accepted. Such restrictions do not subtract the significance of the findings but indicate the possibilities of the further study such as adaptive temporal modelling, cross-domain verification, and extensions that think about fairness.

To summarize, this thesis shows that in early stages of purchase intent could be predicted in a

feasible and extremely useful way as long as thoroughly engineering feature and overall explainability analysis. The study proposes the intuitively-minded and transparent and scalable framework, which enhances the academic research and usability in e-commerce analytics. The paper seeks to provide a platform to future investigations on being able to create a balance between performance, understandability, and applied significance in predictive modelling systems.

Chapter 6

Conclusion and Future work

6.1 Conclusion

Due to the widespread adoption of digital platforms, e-commerce has evolved into a data-intensive industry. The strategic requirement and technical issue are to determine the behavior of customers. Each user interaction such as viewing products, navigating categories, or comparing prices generates valuable behavioral signals. To transform these raw clickstream events into correct predictions of whether an individual desires to purchase something though you must plan, model and work towards fairness and understanding.

This thesis aimed to address a significant gap in the existing studies of e-commerce analytics: the tendency to focus on predictive accuracy at the sacrifice of explainability, realism and methodological rigor. This paper brought the concept of reposition purchase prediction as a decision-support task rather than merely a technical classification problem but a decision-support task whereby model output is expected to be accurate, intelligible and grounded on realistic assumptions concerning the data available.

It goes back to the initial objectives of the research, reviews the methodological contributions, critically considers the findings of the research and places the results within a larger academic and industrial context. It also discusses justness, morality and clarify ability. It also discusses the conditions of the data and modelling decisions and provides understandable research and recommendations on future studies and practice.

6.2 Summary of Research Objectives

The primary objective of the study was to develop an easily understandable and a sound method of anticipating the probability that an e-commerce user session will result in a purchase. The focus of this thesis was realistic prediction where the only information used was available before the purchase decision was made. This is as opposed to the many studies that were carried out earlier using post-hoc data of session completion or the use of recorded buying events.
The objectives of the particular research were the following:\

1. To analyze clickstream behavior at both event and session levels in order to identify patterns that distinguish purchasing from non-purchasing sessions.
2. To develop session-based features that indicate user intent interaction, product exploration and ensure that unwanted target leakage does not occur.
3. To transform the manner in which we imagine purchase prediction into a prefix-based classification problem, which appears like real-time or nearly real-time prediction scenarios.
4. To match the Logistic Regression and LightGBM, both models that are user-friendly and work successfully, on the same terms of the experiment.
5. In order to test models with measures that are good with unbalanced data, it is important

to ensure that results reflect real business performance rather than accuracy only.

6. To apply explainability techniques in order to assess global feature relevance and interpret individual predictions.
7. To examine the concept of fairness, openness, and moral issues, in particular user segmentation and geographic aspects.

6.3 Summary of Methodology

The methodology used in this dissertation was so structured to be systematic, structured and repeatable. The architecture of every step of the pipeline was supposed as a way to support the research objectives and prevent typical issues such as information leaks, overfitting models also non-sensical models.

6.3.1 Data Preparation and Session Construction

The raw click stream data was first cleaned and made equal. Arrangement of timestamps, session identification and normalizing categories were carefully done. Events were inserted into sessions using session identifiers, which ensured chronology of events in time to reflect their natural interrelation among human beings.

A significant methodological decision was to apply prefix-based session snapshots. This implies that a training example is just a fraction of an entire session. This design option has very close resemblance to that of the real life where you are supposed to guess by the end of the session.

6.3.2 Feature Engineering Strategy

The purpose of feature engineering was to transform event data that was both noisy and dimensional into condensed and easy-to-consume session-level features. Features were of three primary kinds:

- Price-related characteristics, such as average, the lowest, the highest, and the range of product prices observed.
- Characteristics of behavioral difference, including the number of various models of products, colours and categories that were employed.
- Categorical dominance features indicate the category, or type of product which most frequently came under consideration within a session.

Above all, any element that had explicitly or implicitly documented the accomplishment of a buy, such as explicit purchase counts or purchase session duration after a purchase, was eliminated. This ensured that the causal and temporal rules of validity were adhered to the letter.

6.3.3 Model Training and Evaluation

Two trained and evaluated models were:

- **Logistic Regression** was chosen because it is easy to understand and see through.
- **LightGBM** was chosen because it can model complicated decision boundaries and non-linear interactions.

The reason why Logistic Regression was selected is that it is simple to study and interpret.
The reason was that LightGBM was selected as able to model complex decision boundaries and non- linear interactions.

Both the models were trained using the same characteristics and ROC-AUC and Precision-Recall AUC were used to test them. The metrics we chose are the large imbalance in the classes in terms of the tasks that are predictive of purchases. The most frequent types of sessions most of the time are the ones that do not result in purchases..

6.4 Discussion of Key Findings

The empirical results offer much information on user behavior and model efficacy.

The two models proved that the behavior of people during the first and middle stages of the session could be used to ascertain the intention of an individual to buy a product even in the absence of the final deal. This helps in the fact that intent signals develop through the course of time based on the patterning of interactions and not a sudden occurrence based on the checkout.

Third, price features were also a complex influence. There was a higher chance of users completing a purchase in case the prices fluctuated a bit within a session. However, when prices were different, people were not so inclined to purchase something. This implies that individuals will easily purchase an item knowing the level of its cost.

The modelling should have started with Logistic Regression since it was easy to understand. LightGBM, in its turn, was more effective in tracing the course of the purchasing sessions. This bartering has a valid point concerning the reality of the business: in many business contexts, it is necessary to identify high-intent users more than to reduce false positives.

6.5 Fairness and Explainability Considerations

The strength of this study is that it is an explainability research. Logistic Regression coefficients gave you the specific effect of features on prediction, whereas SHAP values of LightGBM allowed seeing predictions at a global and local level.

The explainability study study revealed that the models employed mostly align with intuitive behavior cues rather than spurious association. This needs to be open in order to get the stakeholders to have confidence in it and act in accordance with the rules and in a moral manner. The fairness of the geographic and categorical influences was evaluated by evaluating how they affected their fairness. Although the sensitive attributes were not explicitly employed, certain items such as country codes may act as the substitutes to demographic characteristics. This raises serious moral concerns involving individualization, discriminating against others and being even discriminatory.

An explainability framework that would be developed within the frame of this thesis provides a ground to carry on fairness auditing which is sequential and would help the practitioners identify and rectify unintentional differences before implementation.

6.6 Limitations Revisited

Though this study has some imperfections that are to be noted despite the fact there are some significant contributions.

To begin with, the dataset does not contain a transactional signal, which in turn has been confirmed by another party thus in Table 3, labeling required careful assumptions due to the absence of verified transactional signals. These findings are not as useful beyond the study, despite efforts having been made to safeguard the methods employed.

Second, it did not focus its analysis on in-depth sequential modelling, but rather on aggregated session characteristics. This could be easier to comprehend, although it may have omitted minimal changes over time.

Third, the test had been done in an internet-free place. Things may not work the same way in the real world as people use things differently over time, the seasons change, and concepts change.

Finally, the fairness analysis was more of a discovery than a research. The formal fairness metrics and long-term impact assessment was not explored in this thesis, but the specified areas are important in the future research.

6.7 Future Scope

Methodologically, sequence-conscious models like transformar architectures or recurrent neural networks could be studied in order to explain more complicated temporal patterns.

Such models may be effective when strategies to simplify things to comprehend are in play to establish a satisfactory performance to openness ratio.

Systems-wise, the inclusion of the predictive model into a real-time decision engine would allow customizing, targeting interventions and altering recommendation approaches in real-time.

Further studies must adopt causal inference models to distinguish between correlation and causation, which further improve the quality of the future business decision.

Finally, fairness and ethical considerations must be brought more to the general through formal metrics of measurement of bias, counterfactual explanations and user-oriented studies.

6.8 Conclusion

The study was conducted to determine the validity of the prediction of user purchase intent through an e-commerce setting through the usage of clickstream data, with consideration of methodological rigor, interpretability, and practical value. The thesis shows that even without explicit indicators of transaction, the purchase intent can be determined using the session-based behavioural pattern by building a carefully designed modelling pipeline and thoroughly evaluating it using a behavioural pattern.

The outcomes of the experiment give credible results in support of this conclusion. The ROC- AUC

and the PR-AUC of the Logistic Regression model were 0.83 and 0.84 respectively. LightGBM model was associated with a ROC-AUC of 0.88 and PR-AUC of 0.82. Of particular importance is the fact that these results can be obtained with an inherent class imbalance and the only information that can be obtained is partial session information. Precision–Recall AUC is particularly appropriate for classification tasks involving class imbalance and indicated that both models are effective in locating sessions where individuals purchase things. The overall ranking performance of Logistic Regression is also slightly higher, though LightGBM is more able to recall positive (purchase) cases.

These results are even supported by the classification results. LightGBM was given a score of 0.93 in terms of buying sessions and this implies that it is capable of identifying individuals with high interest. It is highly applicable in the real-life in regards to ads targeted to you or altering suggestions. The Logistic Regression model however, performed quite satisfactorily, and was far more understandable. The alteration of various features in the result was clear. It demonstrates a significant trade-off associated with the modern machine learning systems: the possibility to comprehend the result versus the possibility to give correct prediction.

A large portion of this work has to do with how it solves the problem. The research re-modeled purchase prediction as a prefix-based version of session prediction problem which is highly analogous to the process of how decisions are made in the real-time rather than treating it as a post facto classification problem. This approach does not make any false assumptions, causes reduction in data leakage, ensures that predictions are only made on information that is available at the time of making. The metrics given are not influenced by our knowledge of future, and hence include a better way to view model performance.

The feature-level analysis and explainability make one notice massive behavior patterns that go beyond the current dataset being analyzed. Predictors that included characteristics that quantified the level of exploration of products were some of the best predictors and this could be the number of different models of clothing looked at during a session. A large degree of diversity was always associated with reducing possibility of purchase and this may imply that the individual was hesitant or simply window shopping. On the contrary, focused patterns of interaction were also a sign of increased probability of purchase. The features associated with price also served to demonstrate that the user who exhibits the steady price-related engagement has higher likelihood of converting which is an additional finding to the extant body of e- commerce behavior literature.

This thesis contributes to the growing body of research advocating explainable and accountable machine learning in critical business applications.This paper shows that systems that are high performing can be understood by combining understandable baseline models with understandable ensemble approaches. Coefficient analysis and SHAP-based explanations can be used to check, confirm, and explain model decisions to non-technical people. This is gaining prominence in business as well as the government arena.

The outcomes are encouraging and this research demonstrates the difficulty of anticipating an intention to purchase. It has much of the unseen that influences the way individuals behave such as altering their plans, external world and their own preferences. All these things may not be exhibited by the use of the clickstream data alone. The level of consistency in the models per the various metrics of quantification of evaluations indicate that the behavioral signals in the interactions of the session hold a predictive strength.

In this thesis, the study has proved that it is possible and even profitable, to use the information about the clickstream to predict the intention to purchase, though it is necessary to adhere to the execution of the research, apply the right criteria and provide the clarity in the findings. The results have applicability in the context of various online business environments and can be useful as practical information to the practitioners who are interested in developing data-driven decision support systems. It is shown in work done that machine learning research has to be complemented with realistic assumptions, with ethical considerations and realizing model design. They are principles that will remain significant as more and more data-driven personalization is progressive.

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